Phylogenetics

Introduction to molecular dating

RL-V3 MPP

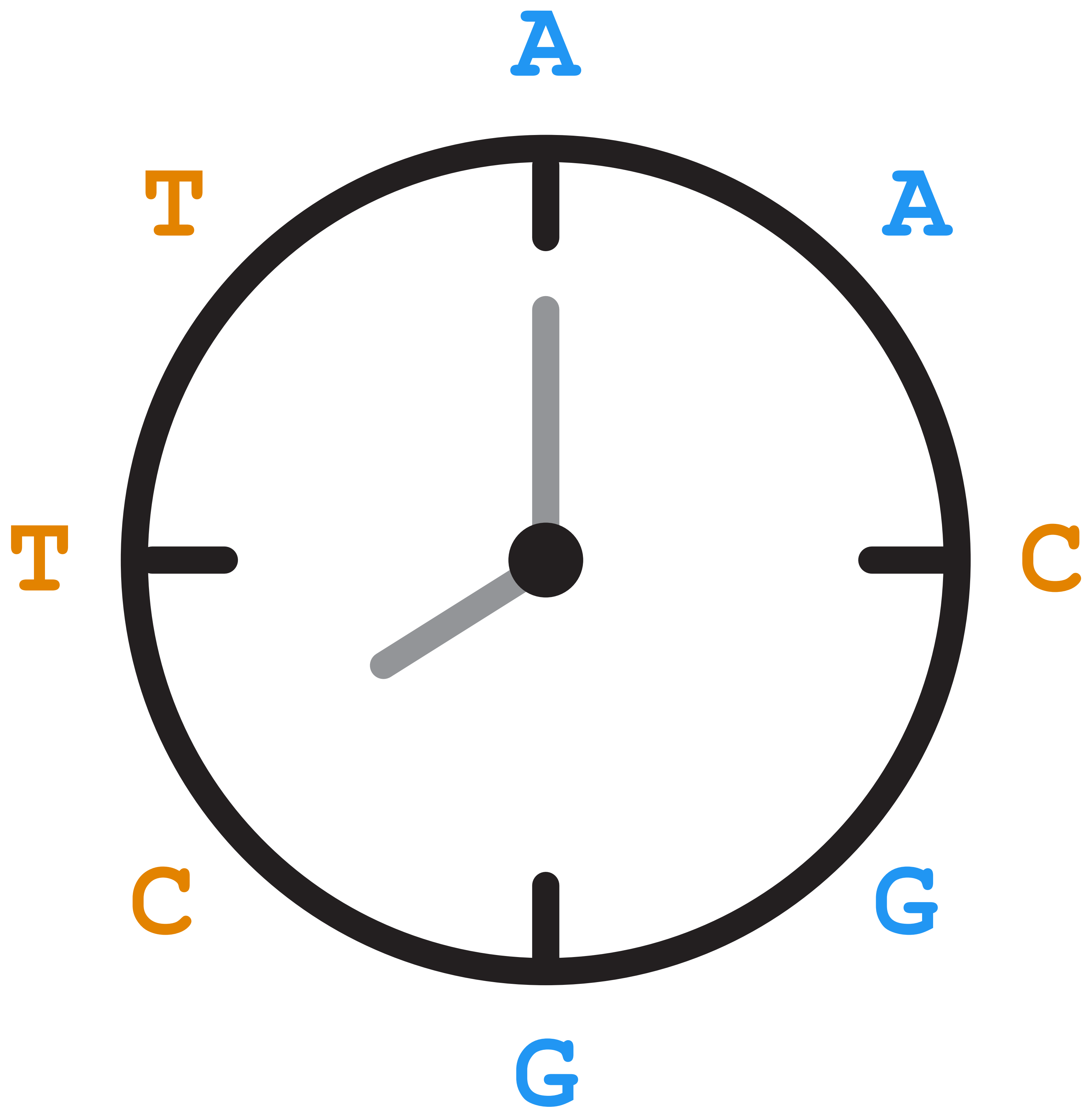
Rachel Warnock

09.04.26

Hypothesis testing?

Today's objectives

- Molecular dating
- Intro to BEAST2



Recap

How do we find the 'best' tree?

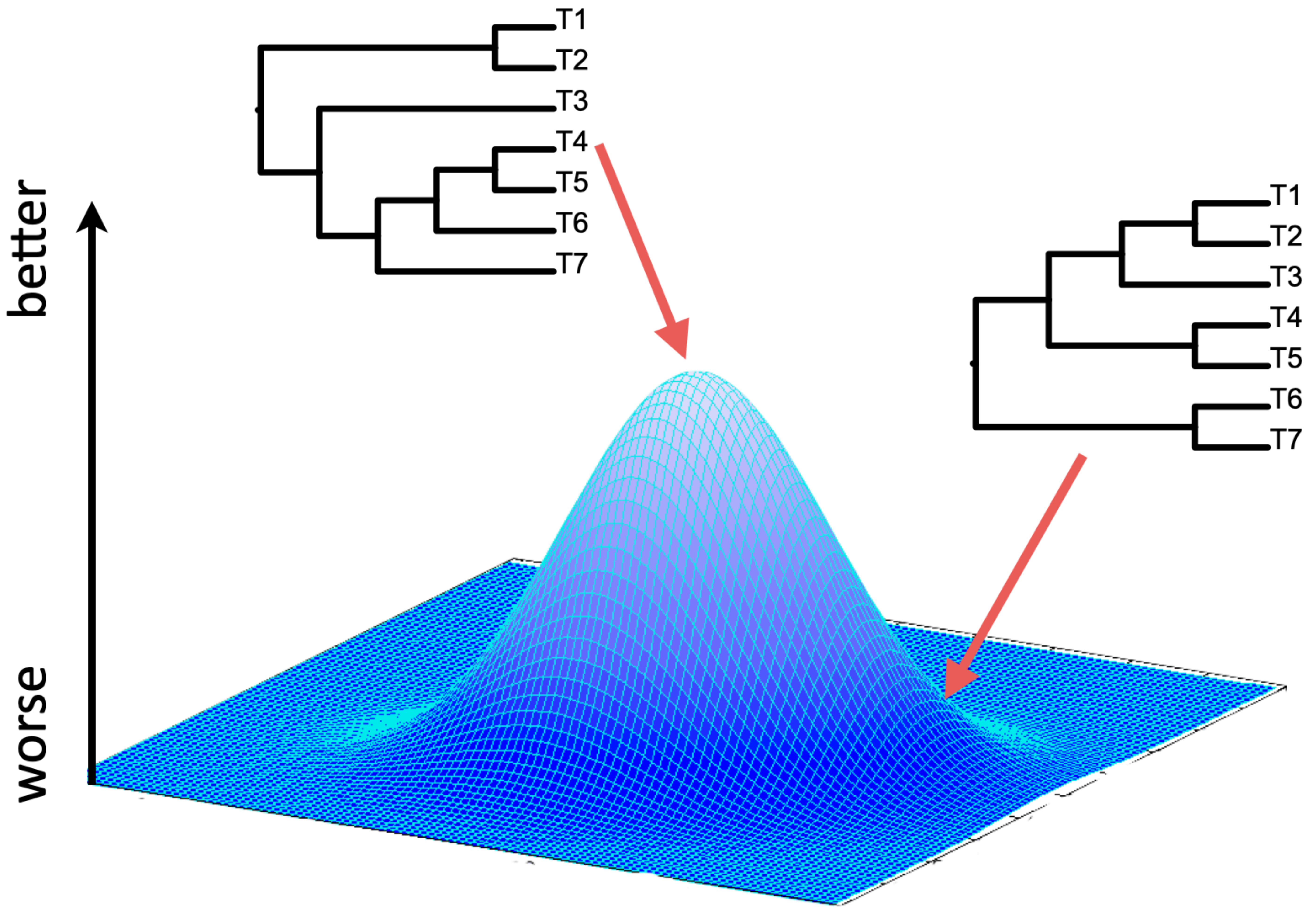


Image source Tracy Heath

It depends how you measure ‘best’

Method	Criterion (tree score)
Maximum parsimony	Minimum number of changes
Maximum likelihood	Likelihood score (probability), optimised over branch lengths and model parameters
Bayesian inference	Posterior probability, integrating over branch lengths and model parameters

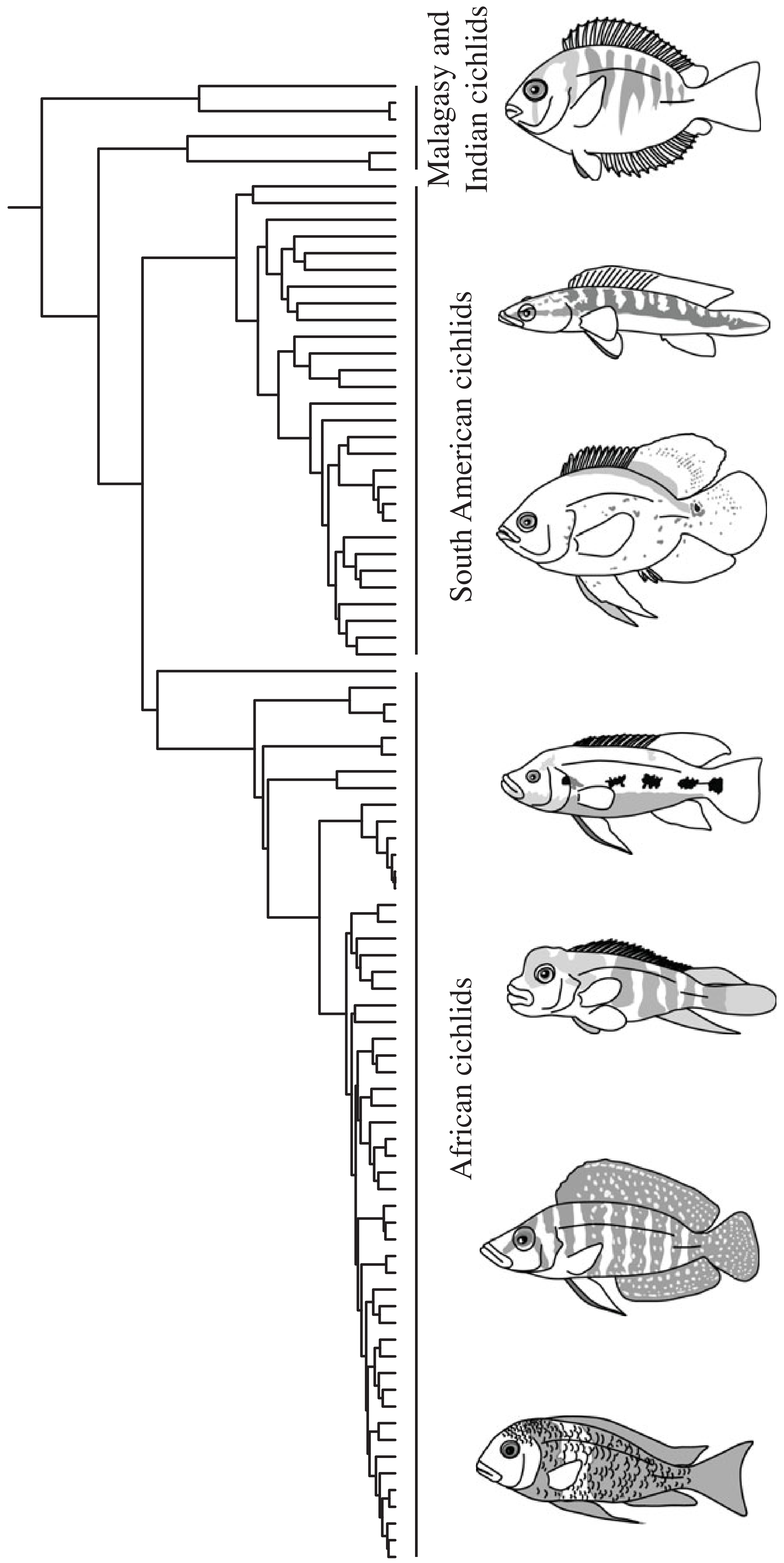
Both maximum likelihood and Bayesian inference are model-based approaches

Note these are not the only approaches to tree-building but they are the most widely used

Introduction to molecular dating

What can we learn from trees?

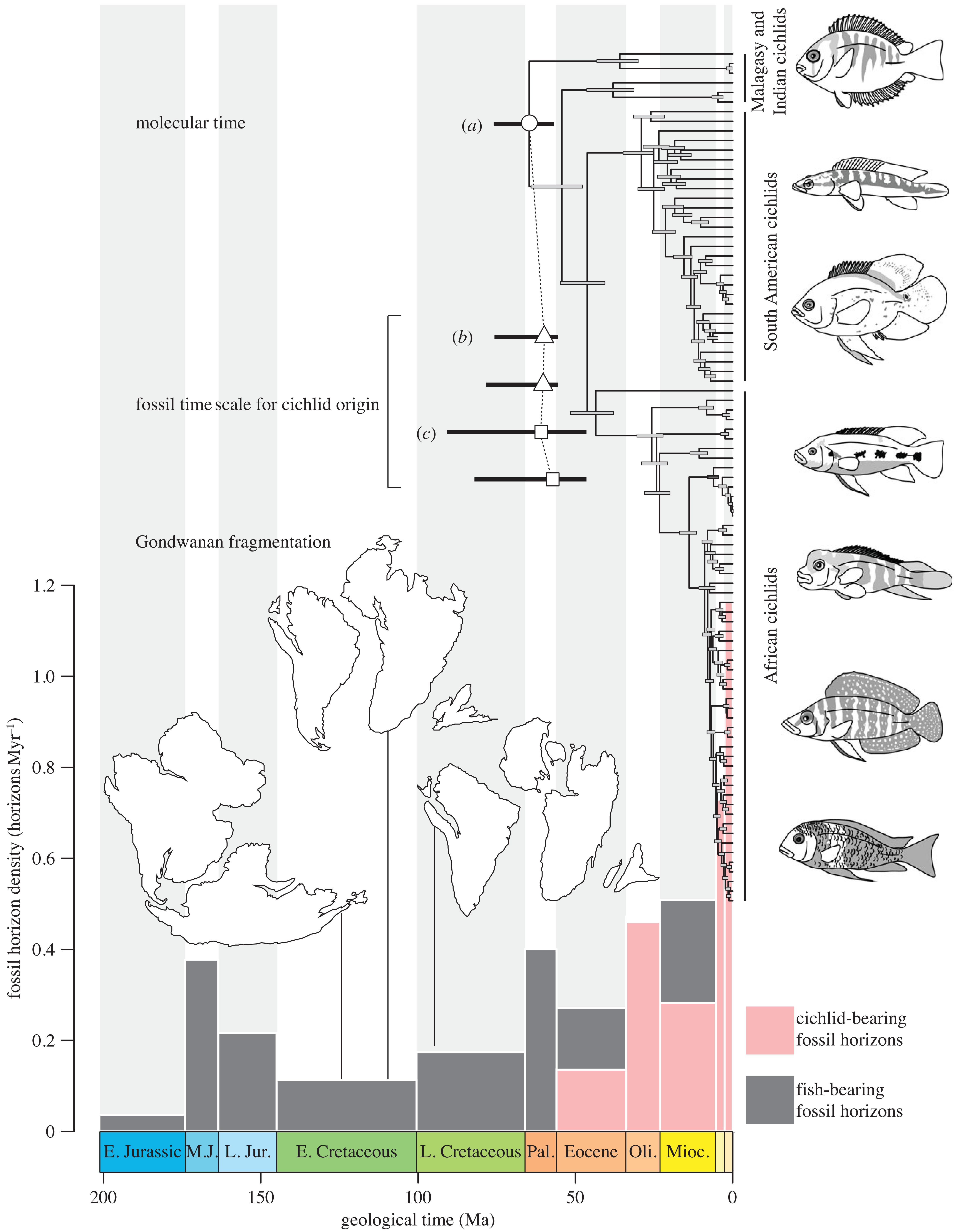
- Evolutionary relationships



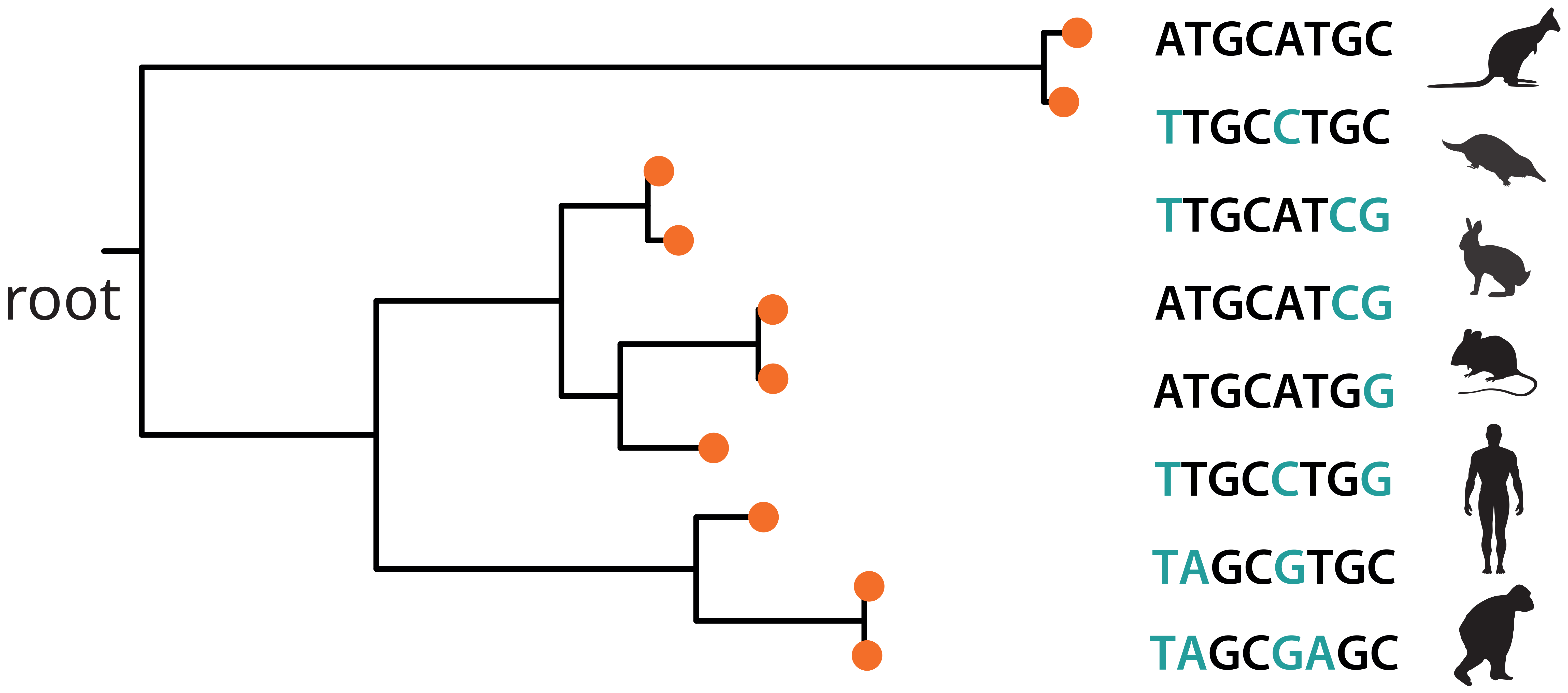
What can we learn from trees?

- Evolutionary relationships
- Timing of diversification events
- Geological context
- Rates of phenotypic evolution
- Diversification rates

Image adapted from *Friedmann et al.* ([2013](2013))



Molecular (or morphological) characters are not independently informative about time

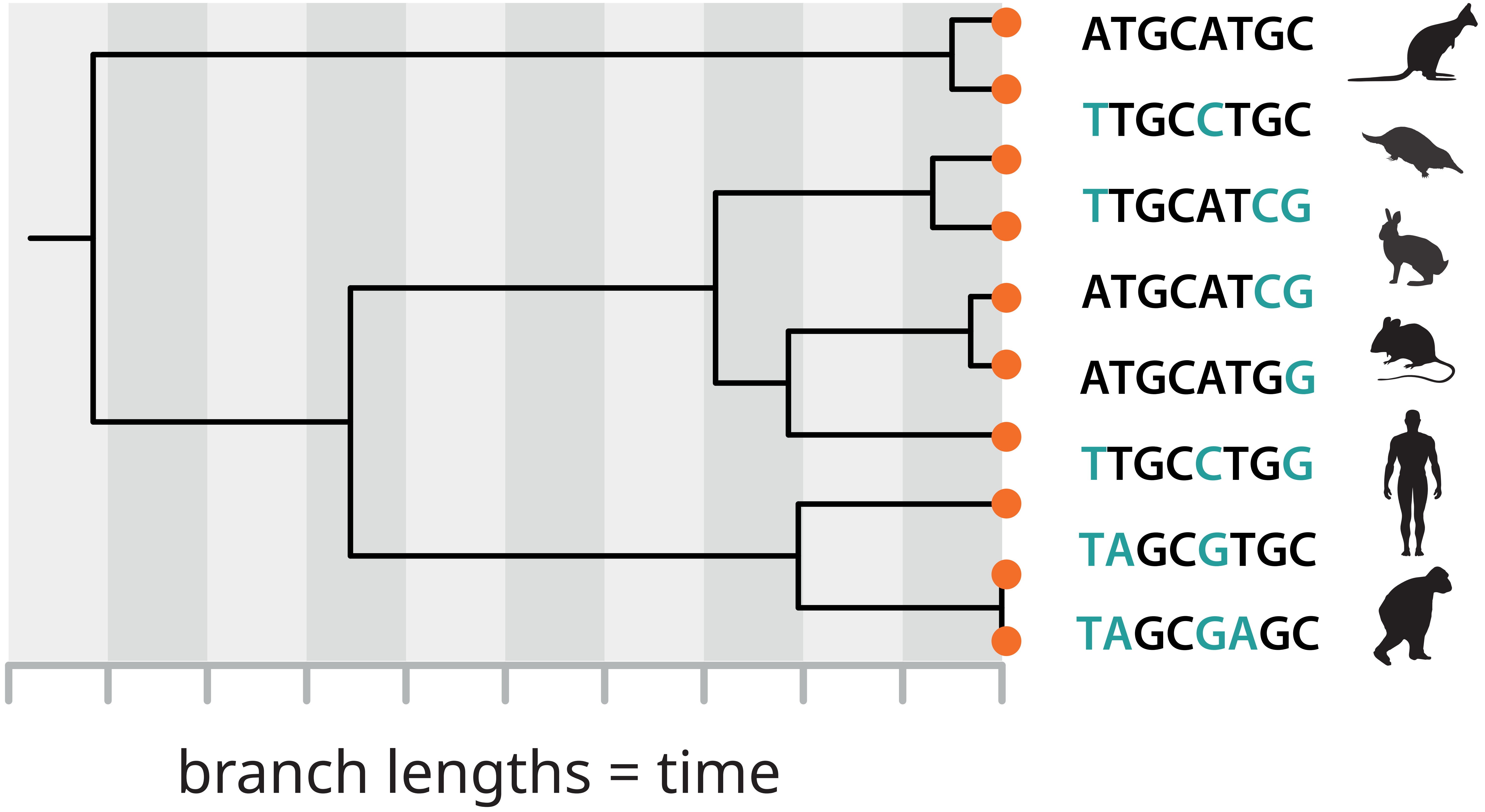


branch lengths = genetic distance
$v = rt$

Slow rate, long interval **or** fast rate, short interval?

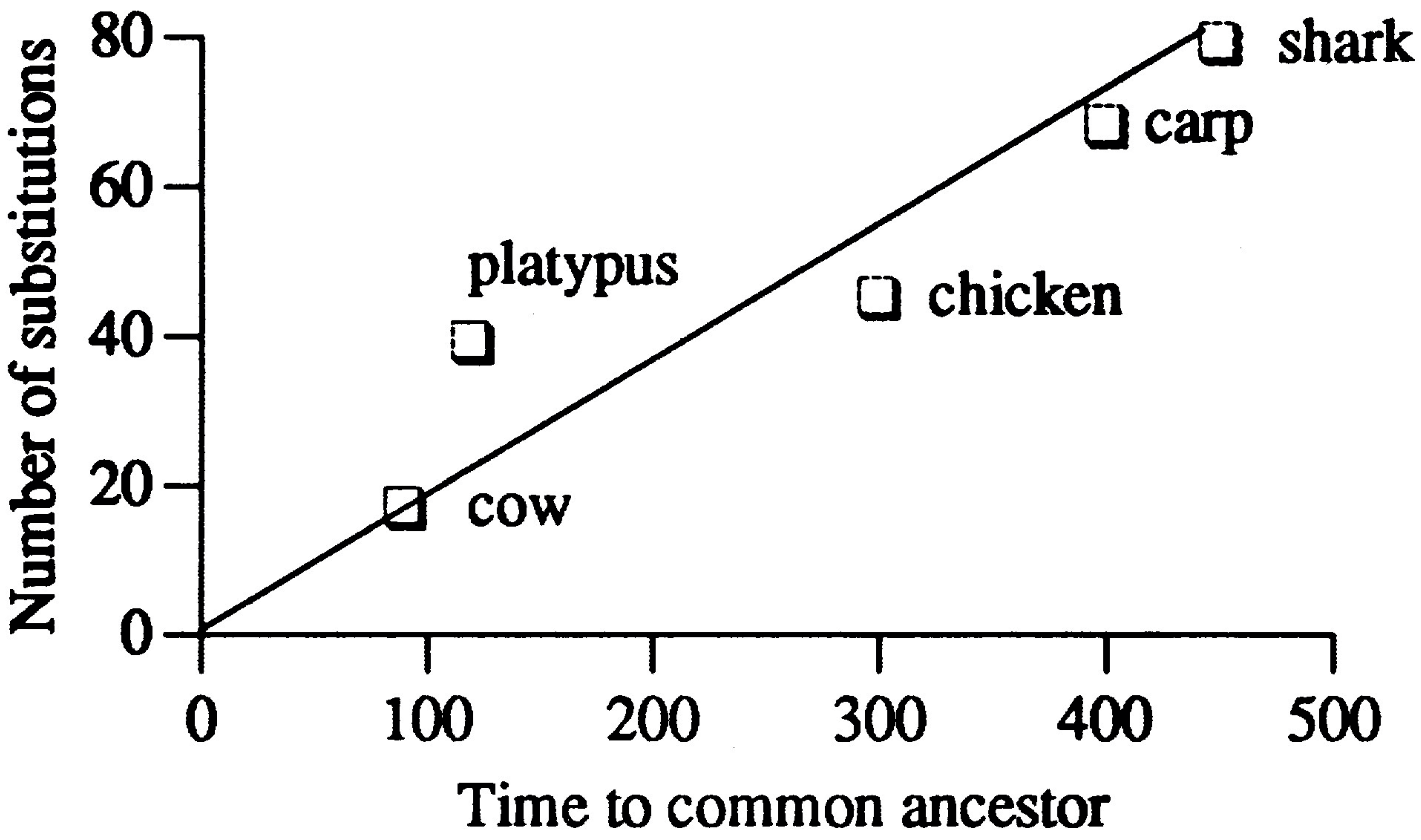
Goal: to disentangle evolutionary rate and time

Molecular (or morphological) characters are not independently informative about time



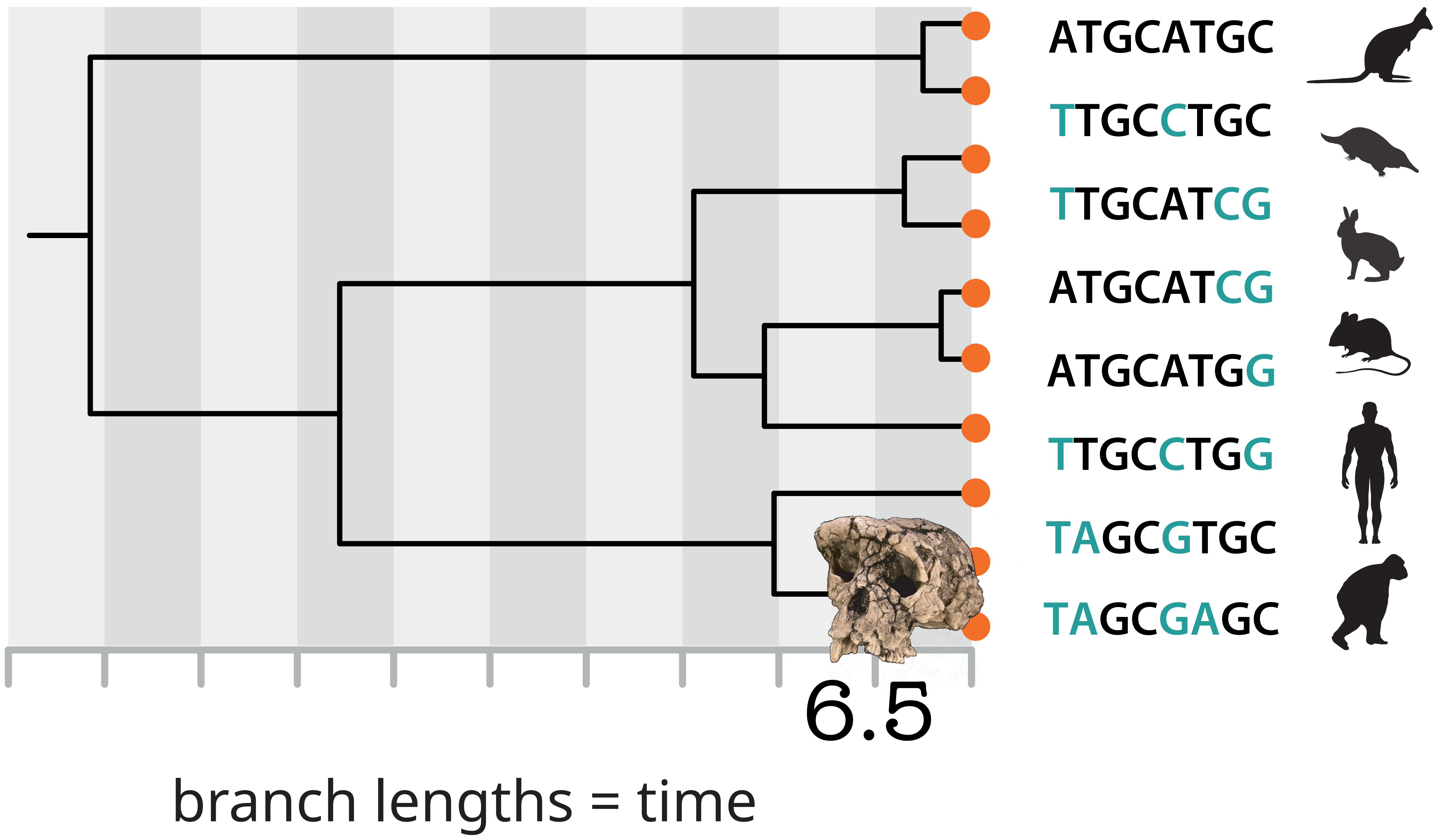
Goal: to disentangle evolutionary rate and time

The molecular clock hypothesis



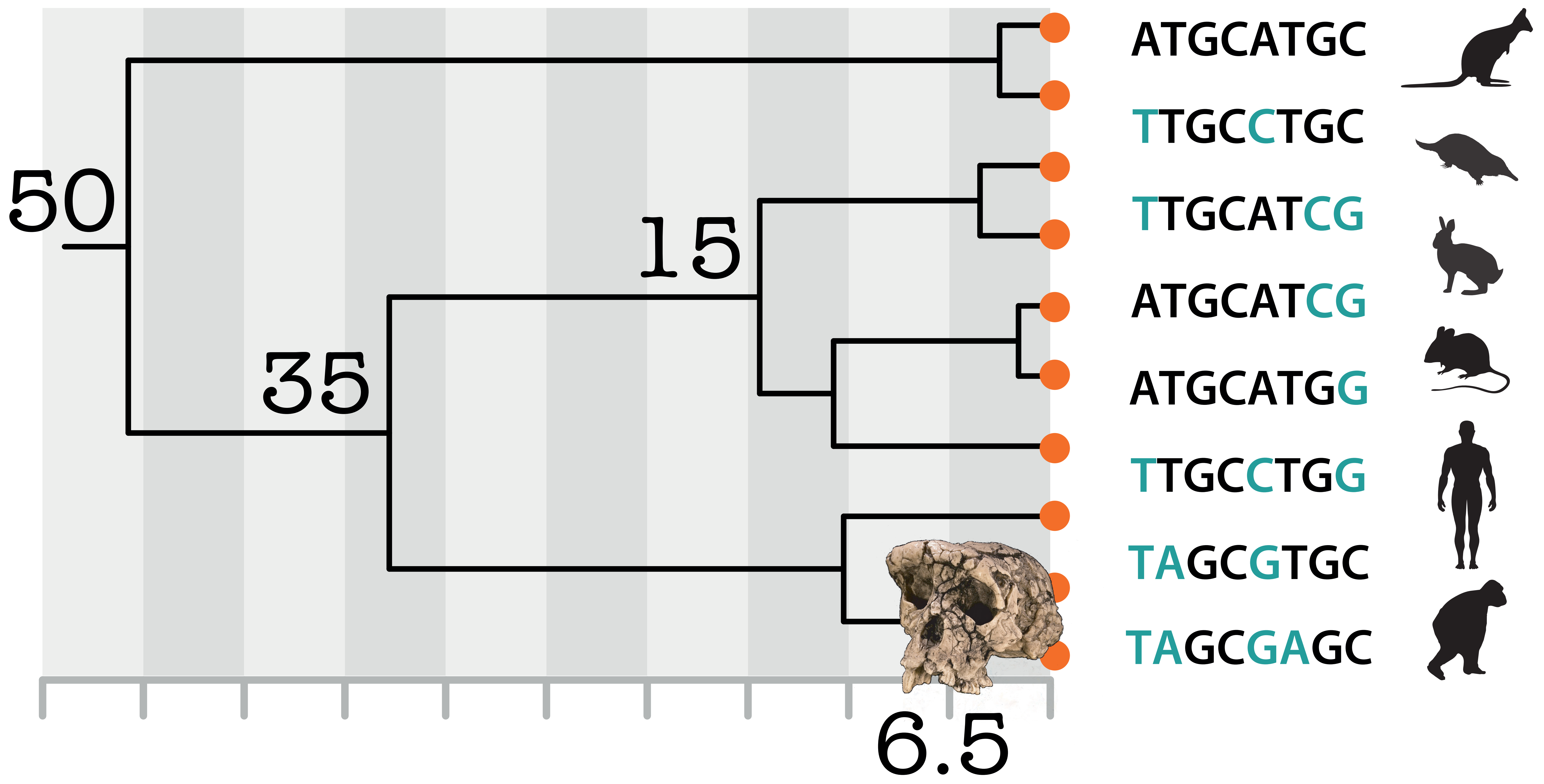
Molecules as documents of evolutionary history *Zuckerkandl & Pauling* ([1965](1965))
A history of the molecular clock *Morgan* ([1998](1998))

Calibrating the substitution rate



Temporal evidence of divergence for one species pair let's us calibrate the average rate of molecular evolution

Calibrating the substitution rate

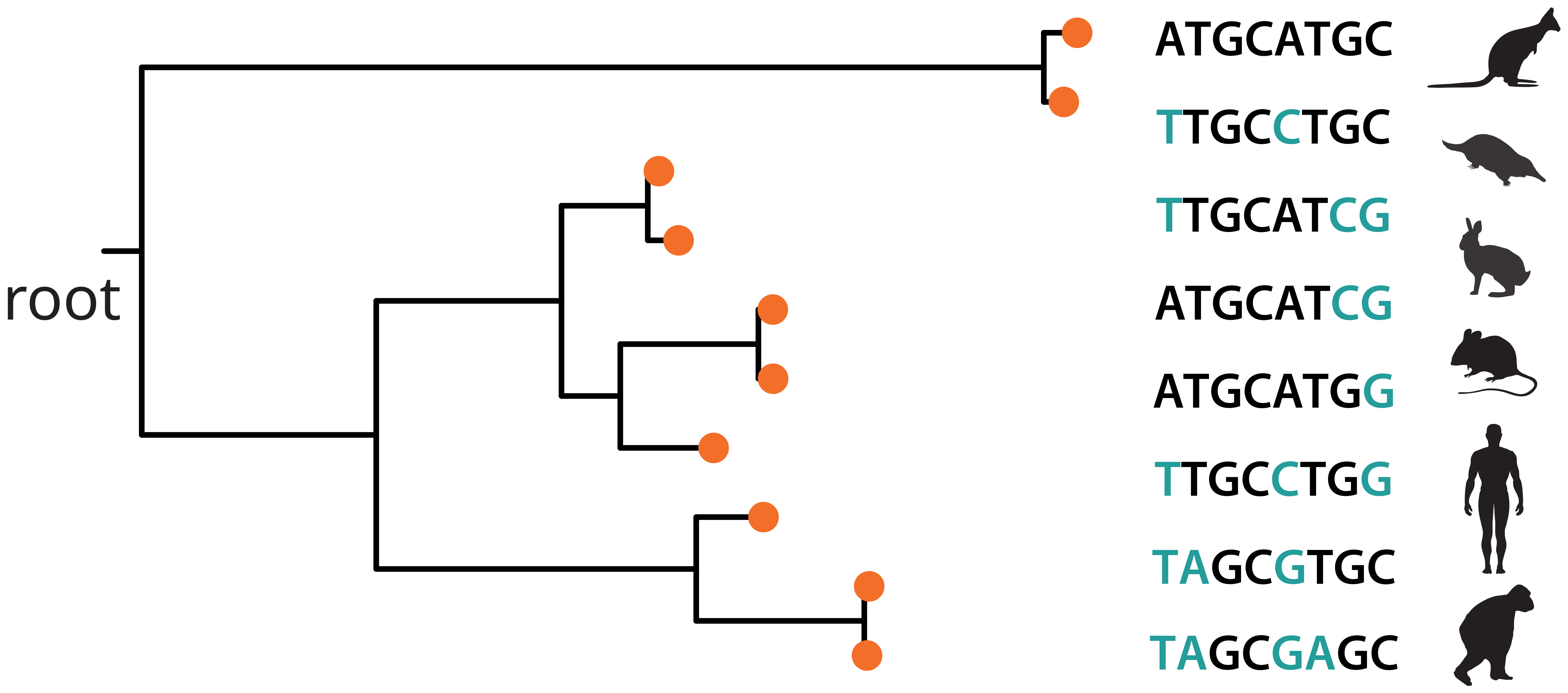


branch lengths = time

We can use this rate to extrapolate the divergence times for other species pairs

Molecular dating: challenges

Rate and time are not **fully identifiable**!

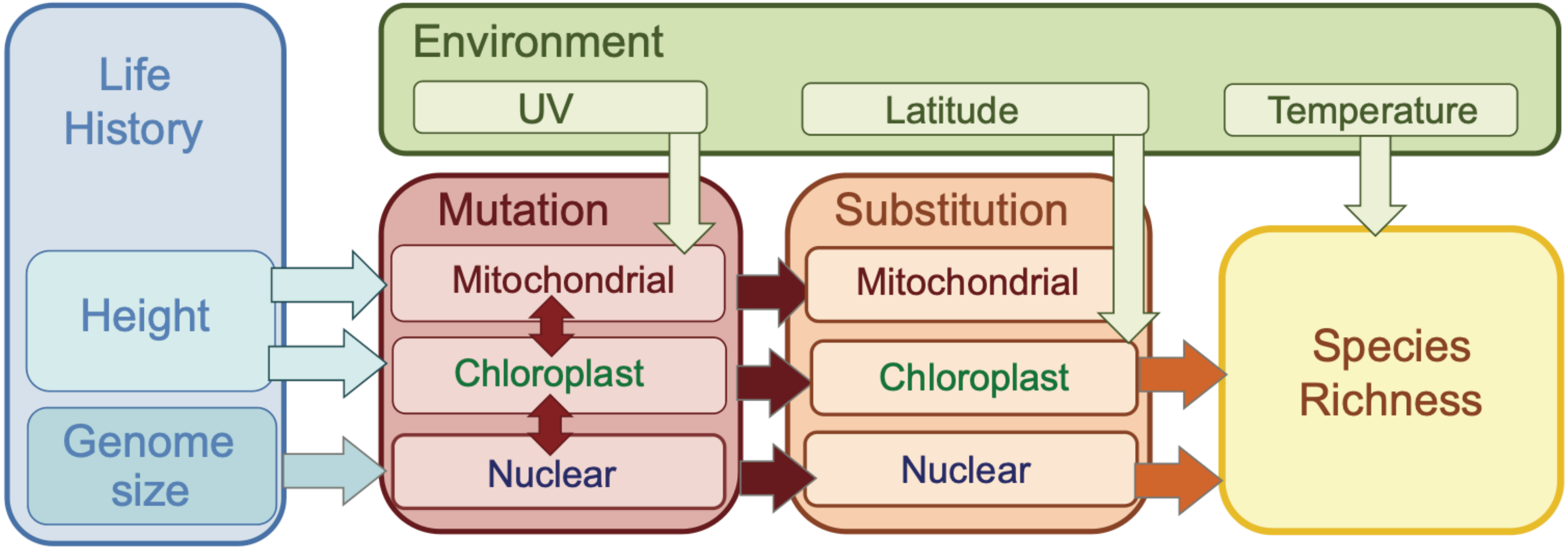


branch lengths = genetic distance

$v = rt$

Molecular dating: challenges

Many variables contribute to **variation in the substitution rate**



Bromham et al. (2015)

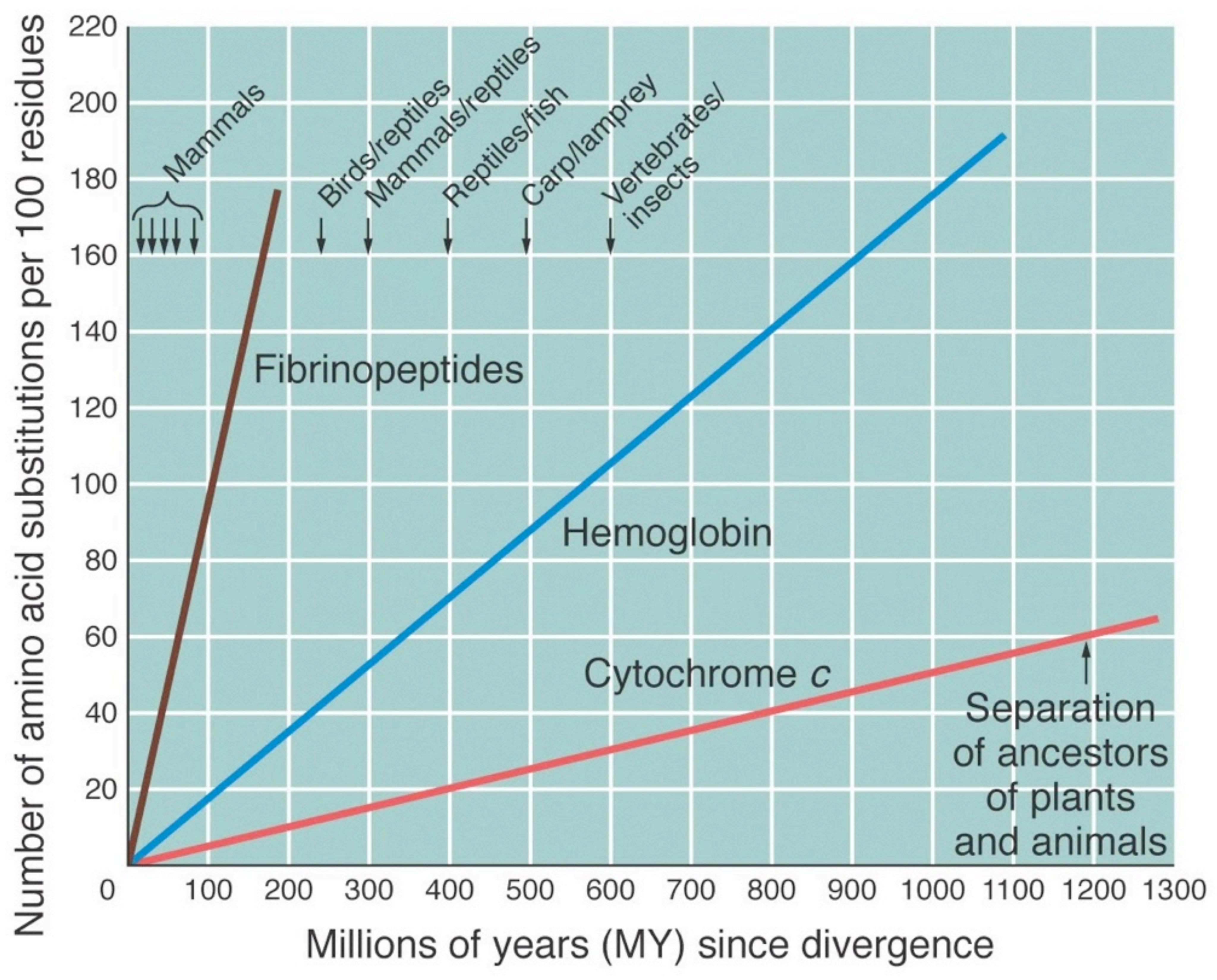
Molecular dating: challenges

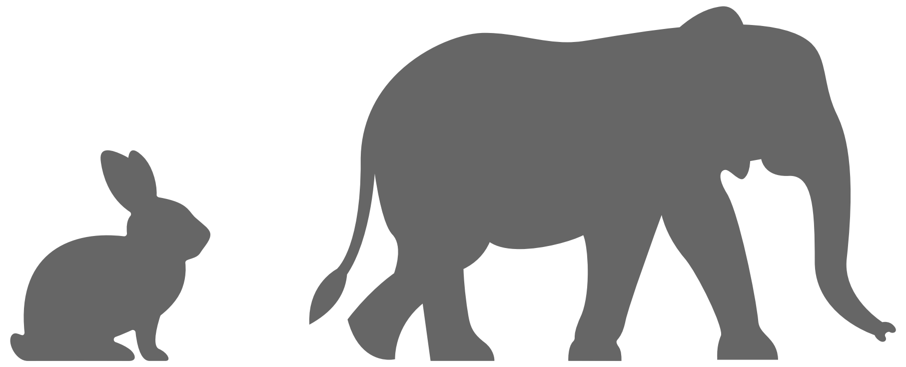
Many variables contribute to **variation in the substitution rate**

The molecular clock is not constant

Rates vary across:

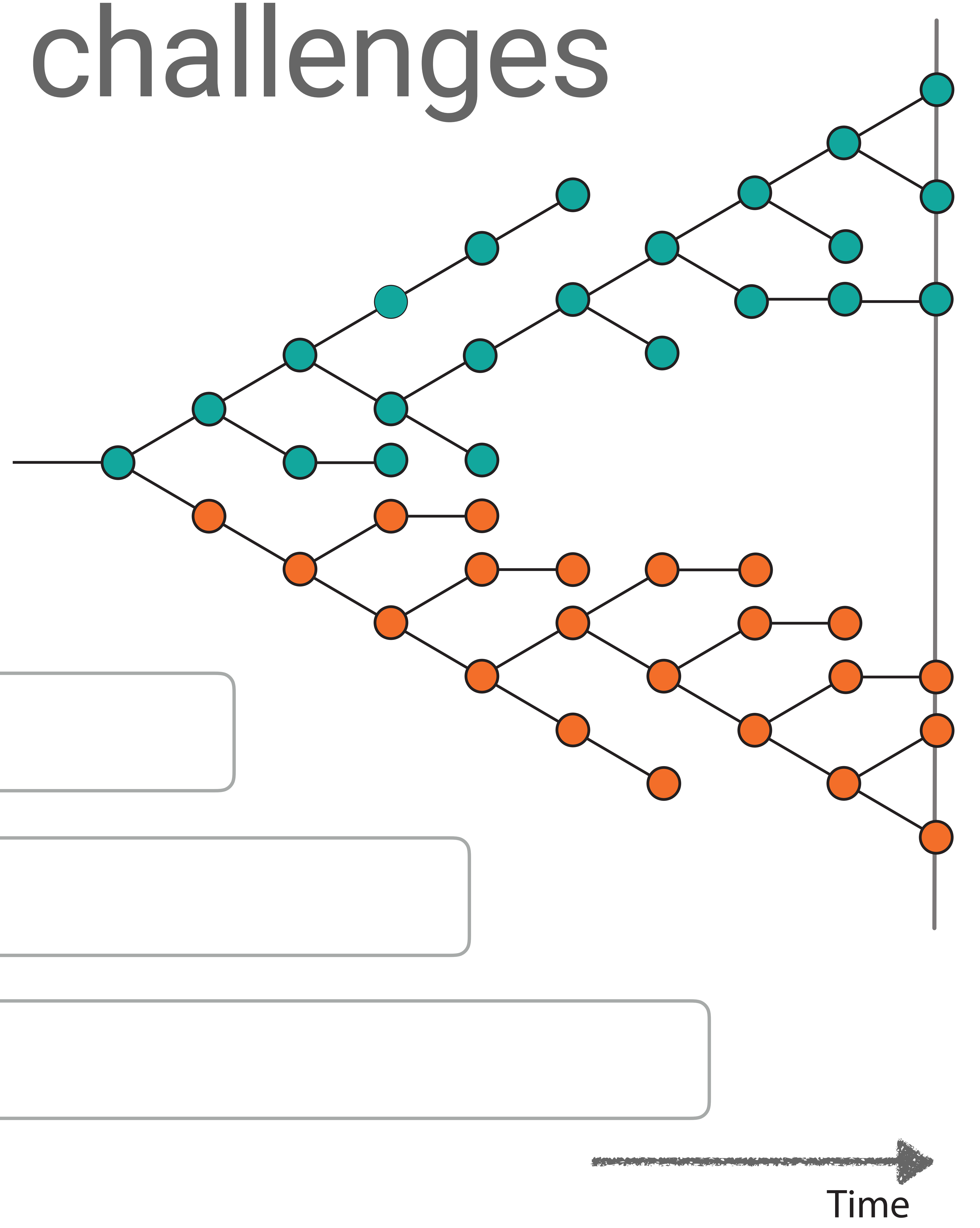
- taxa
- time
- genes
- sites within the same gene





Molecular dating: challenges

Calibrations are imprecise



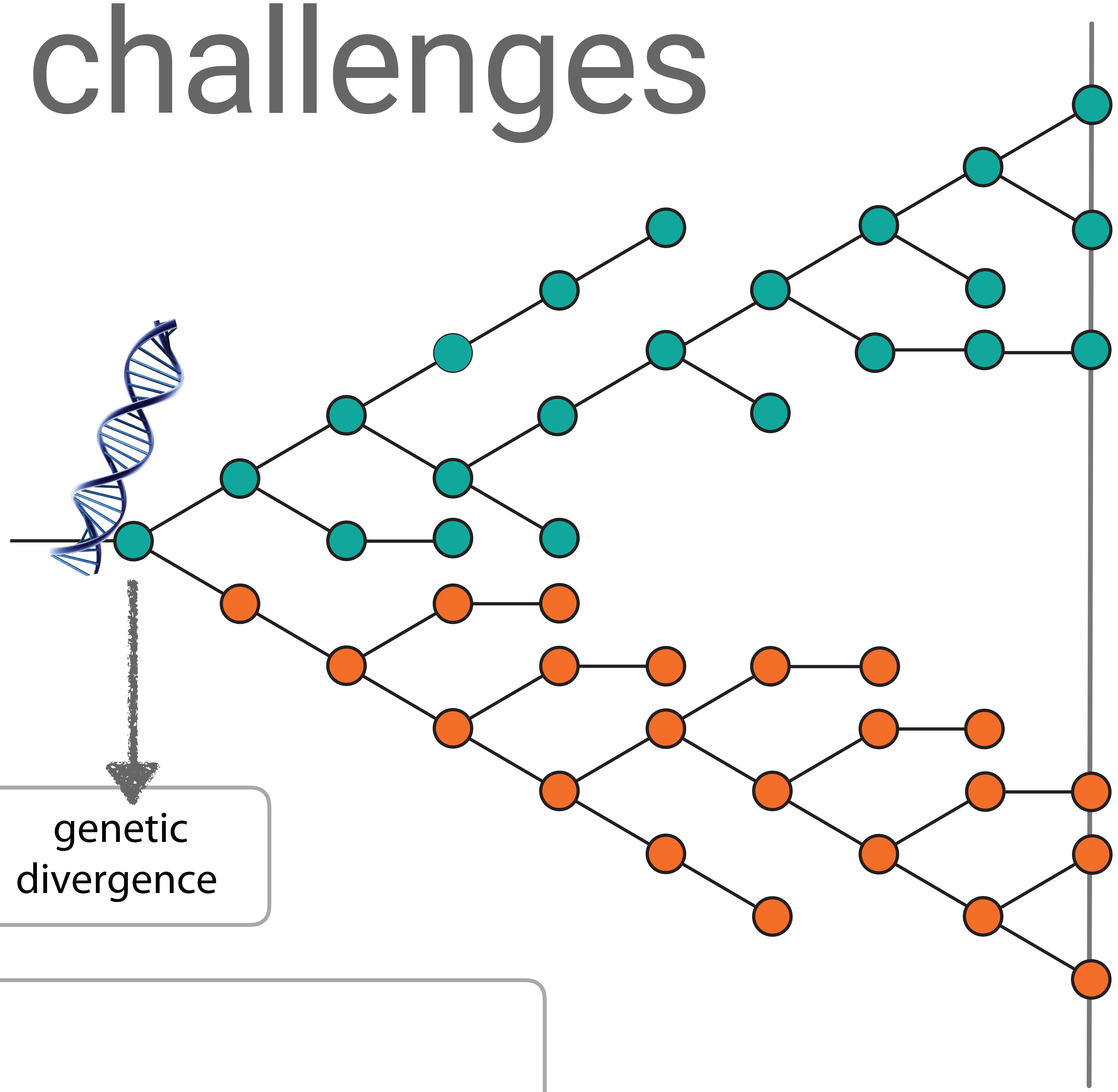
Molecular evolution:

Morphological evolution:

Fossil preservation:

Molecular dating: challenges

Calibrations are imprecise



Molecular evolution: genetic divergence

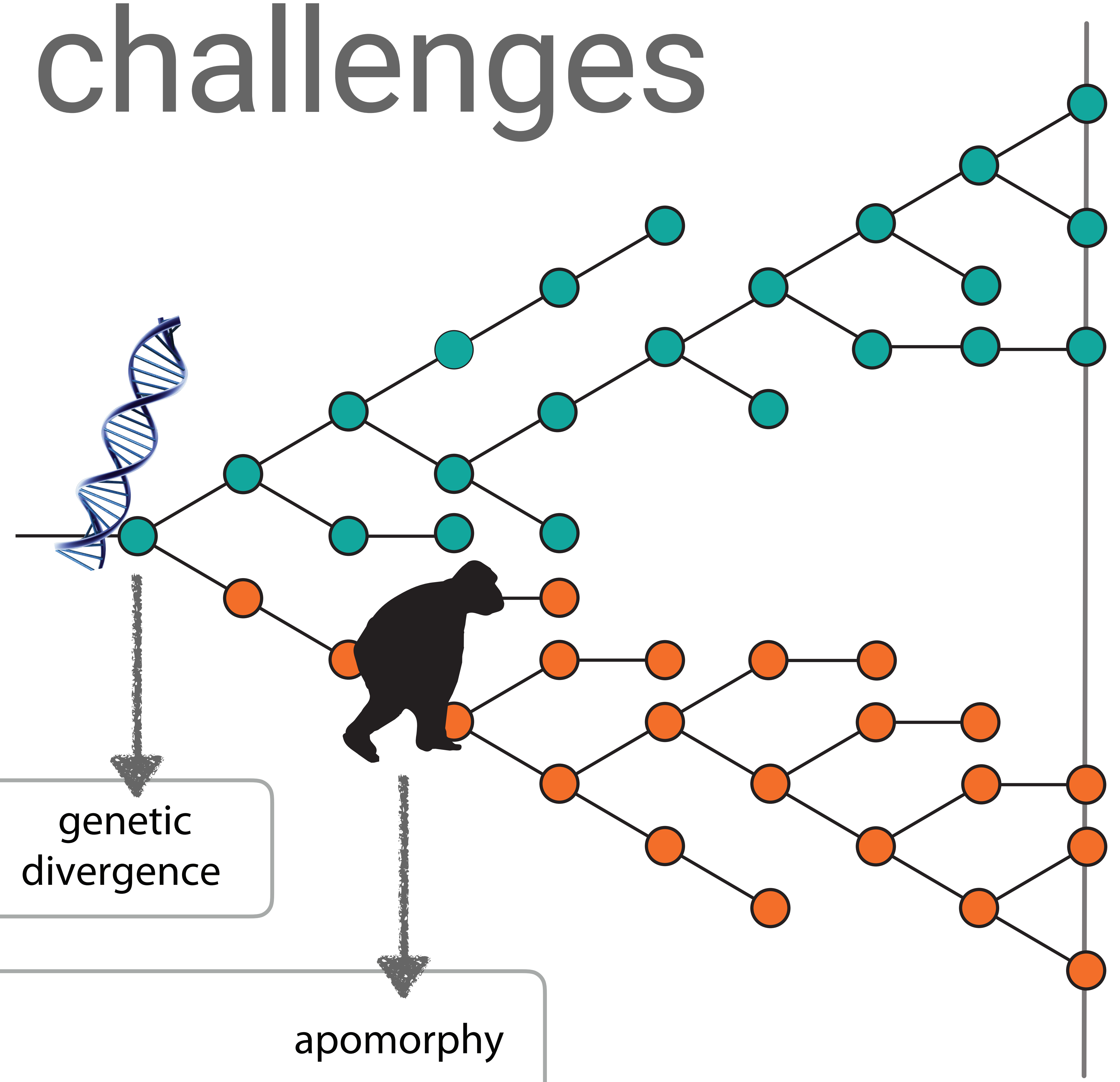
Morphological evolution:

Fossil preservation:

Time

Molecular dating: challenges

Calibrations are imprecise



Molecular evolution: genetic divergence

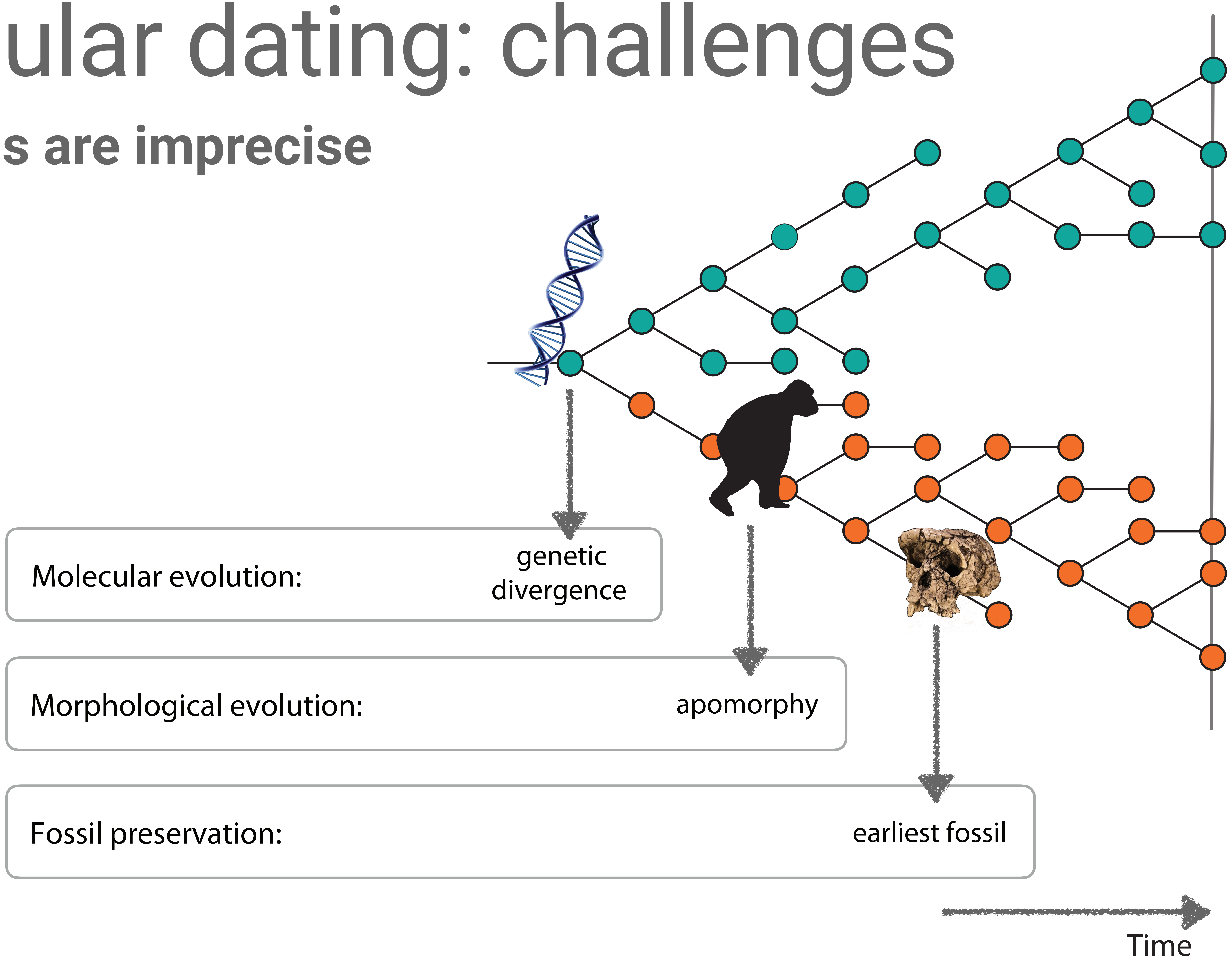
Morphological evolution: apomorphy

Fossil preservation:

Time

Molecular dating: challenges

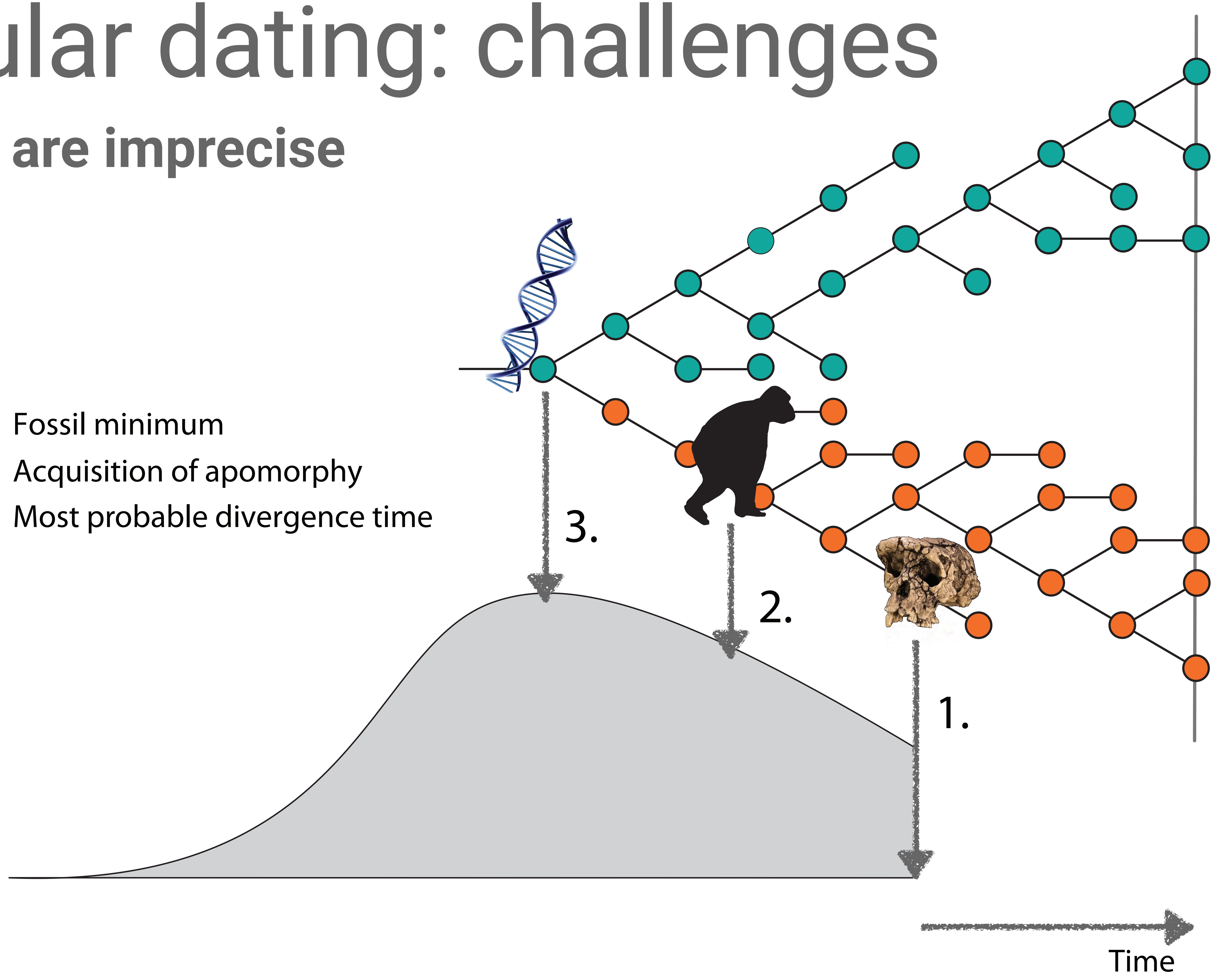
Calibrations are imprecise



Molecular dating: challenges

Calibrations are imprecise

1. Fossil minimum
2. Acquisition of apomorphy
3. Most probable divergence time



Molecular dating: challenges

Summary

1. Rate and time are not fully identifiable

2. The substitution rate varies

3. Calibrations are imprecise

→ we need a flexible statistical framework that deals well with uncertainty!

Bayesian divergence time estimation

We use a Bayesian framework

likelihood

priors

$$P(\text{model} \mid \text{data}) = \frac{P(\text{data} \mid \text{model})\, P(\text{model})}{P(\text{data})}$$

posterior

marginal
probability of the
data

Bayesian divergence time estimation

The data

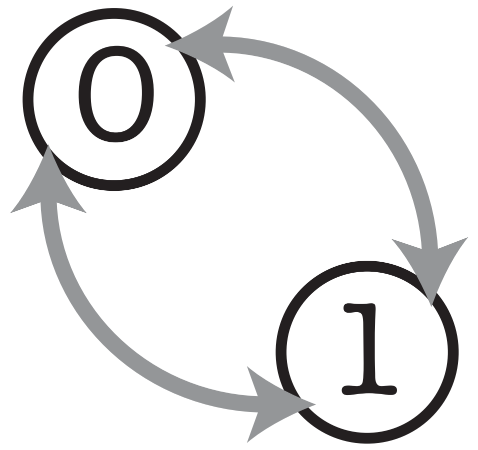
and / or

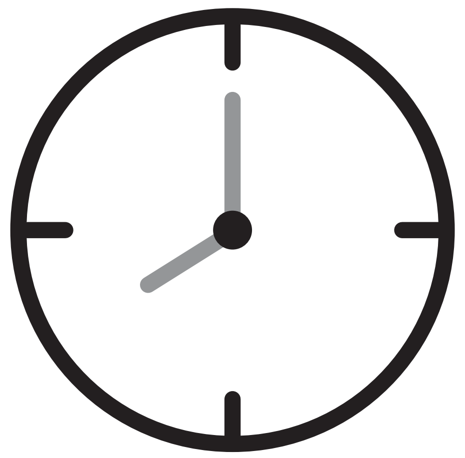
0101... ATTG...
1101... TTGC...
0100... ATTC...

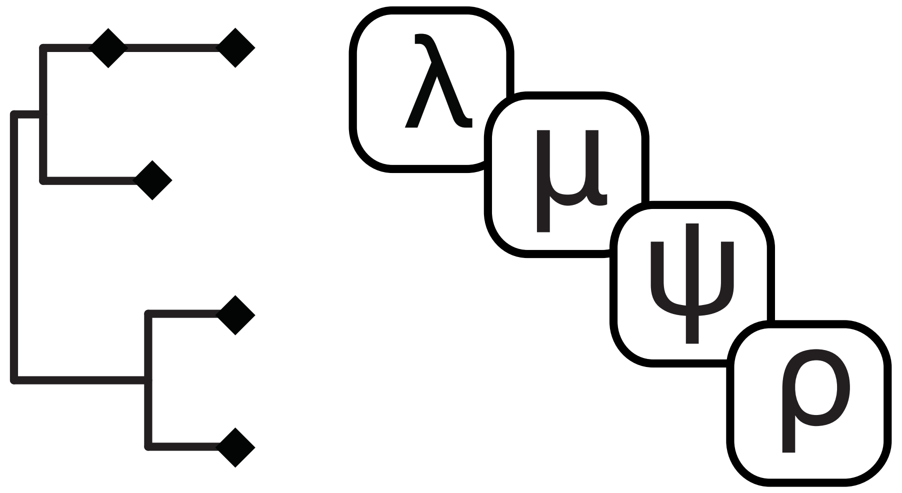
phylogenetics characters

sample ages

3 model components







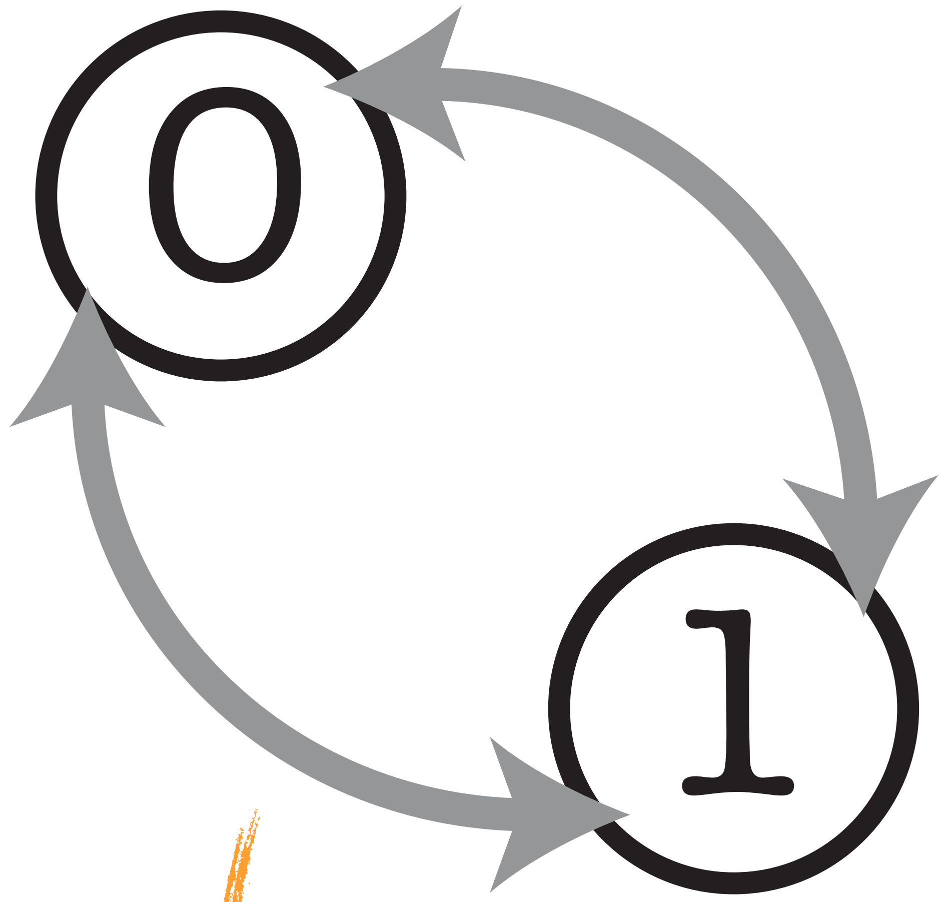
substitution model

clock model

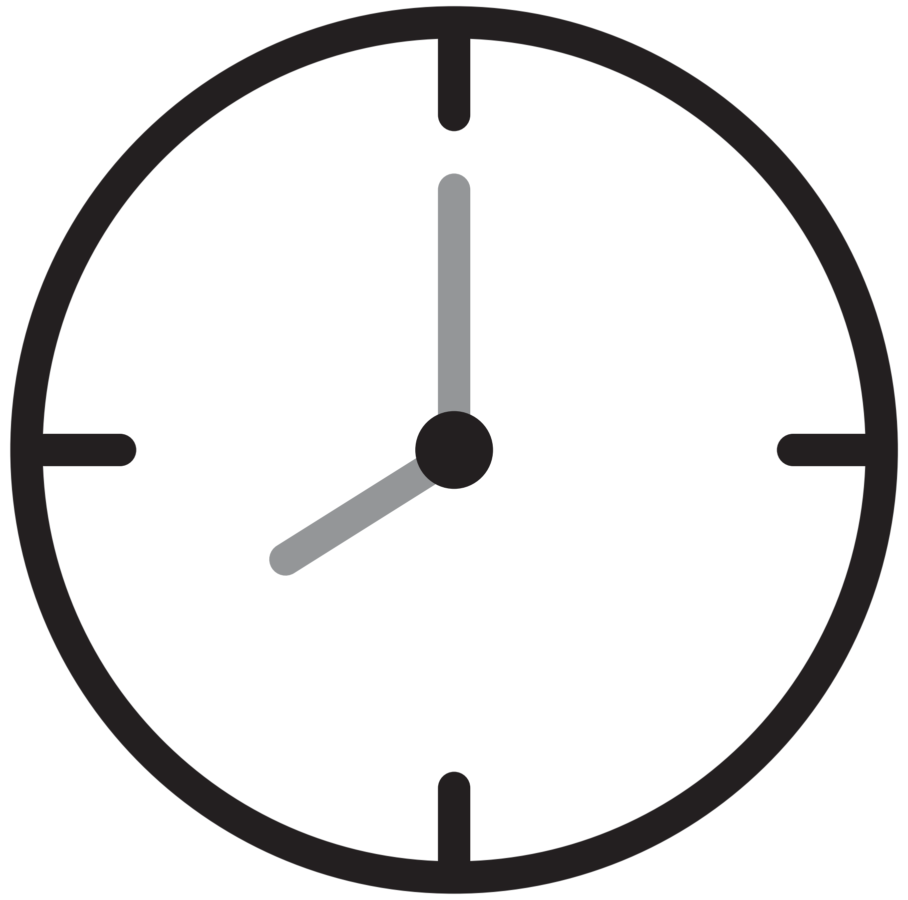
tree and tree model

Understanding the tripartite approach to Bayesian divergence time estimation
Warnock, Wright. (2020)

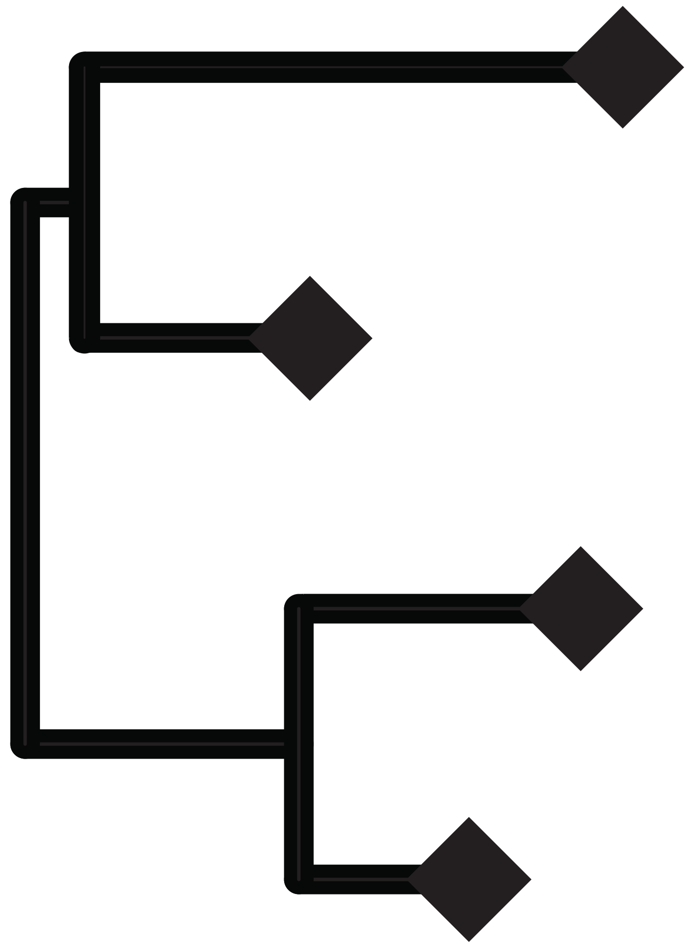
substitution model



clock model

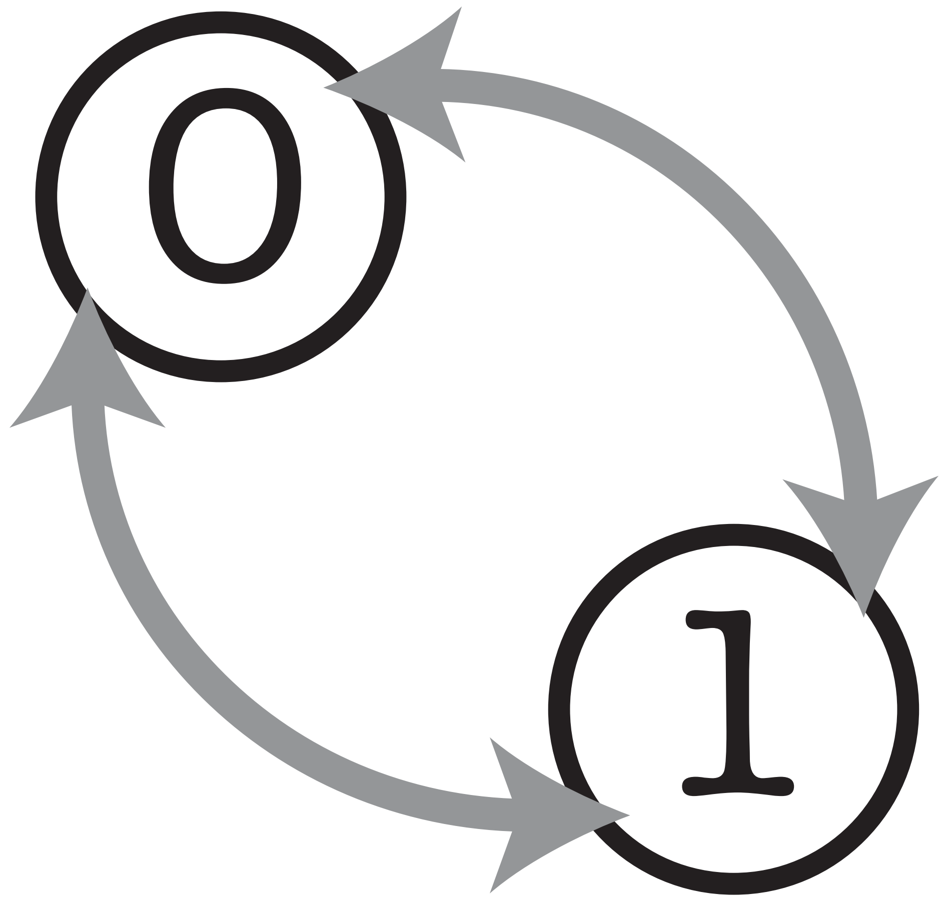


tree model

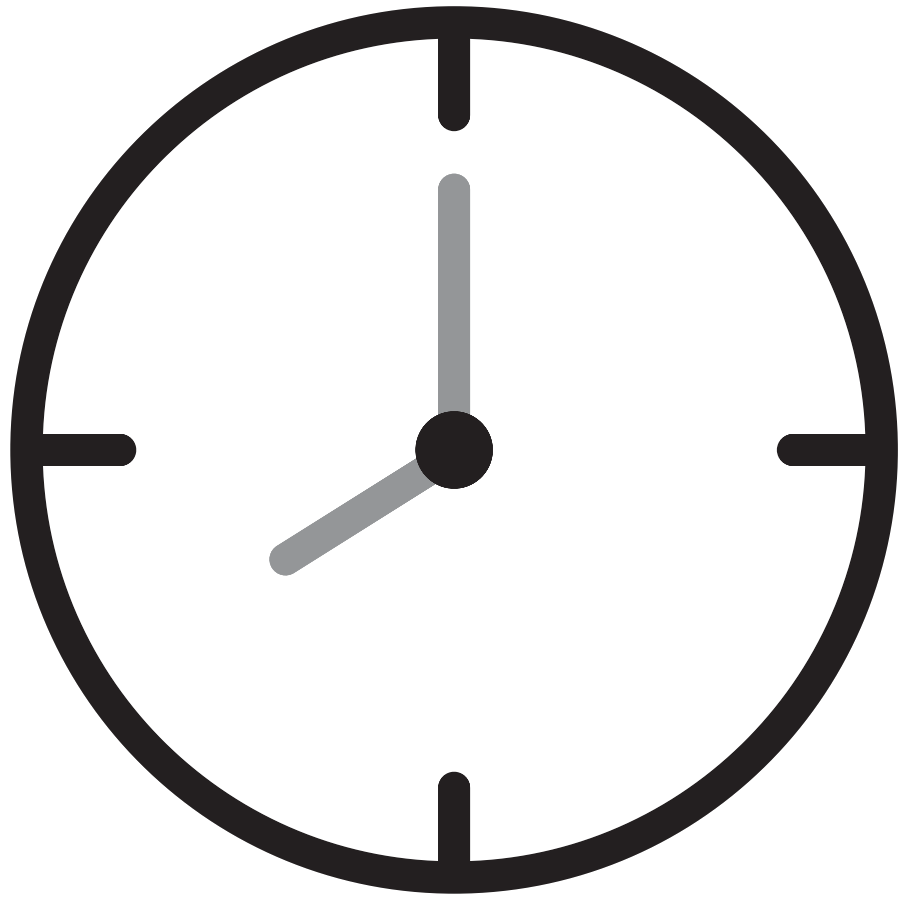


How likely are we to observe a change between character states? e.g., A → T

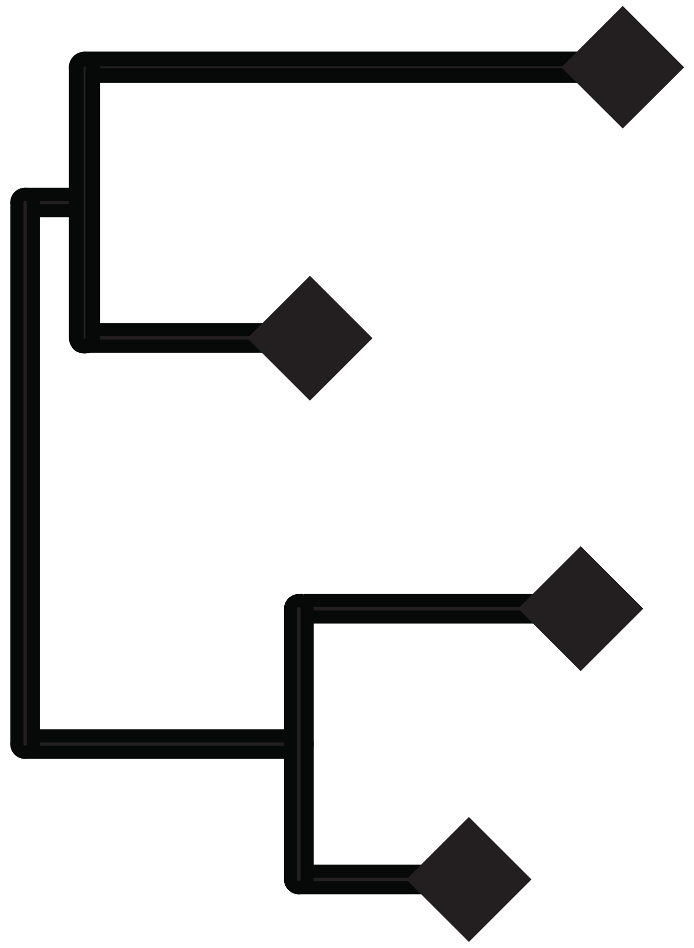
substitution model



clock model

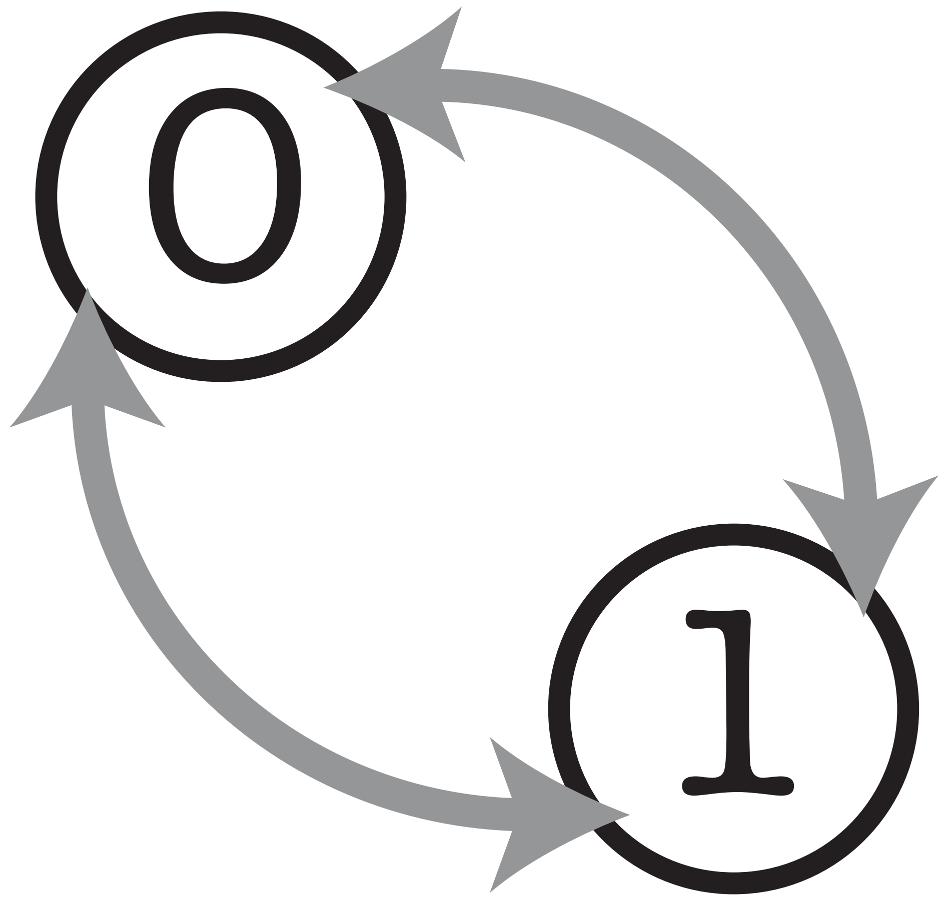


tree model

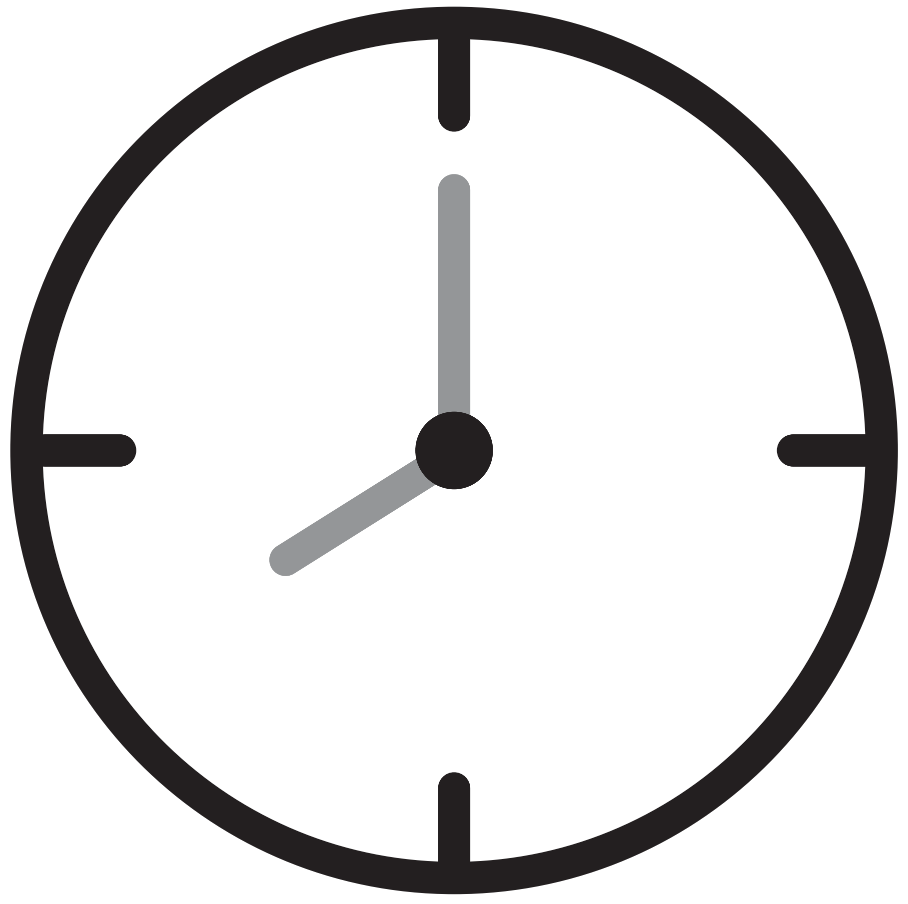


How have rates of evolution varied
(or not) across the tree?

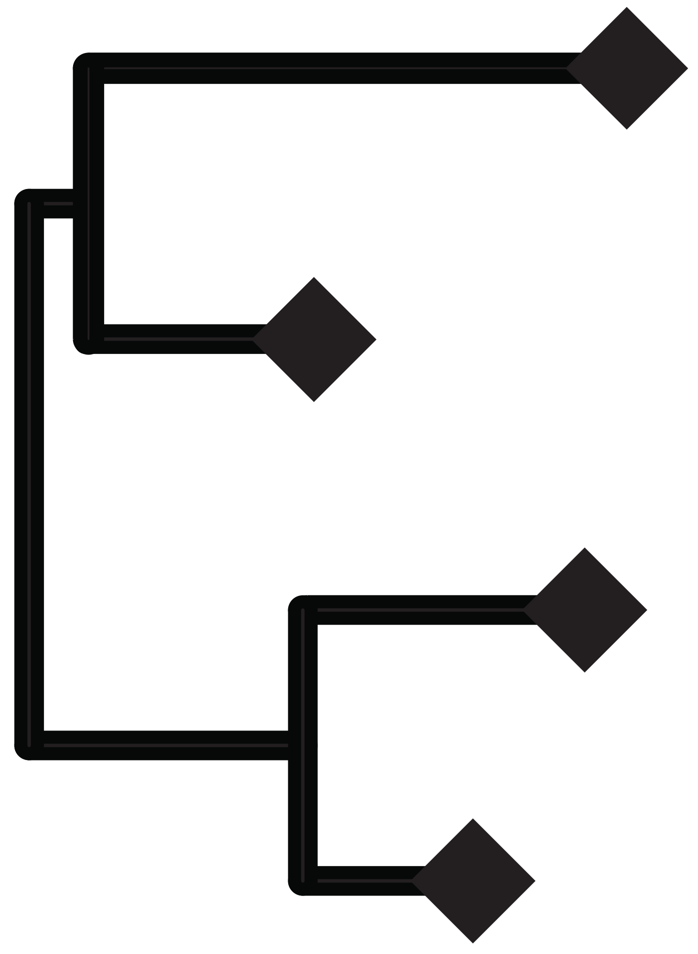
substitution model



clock model

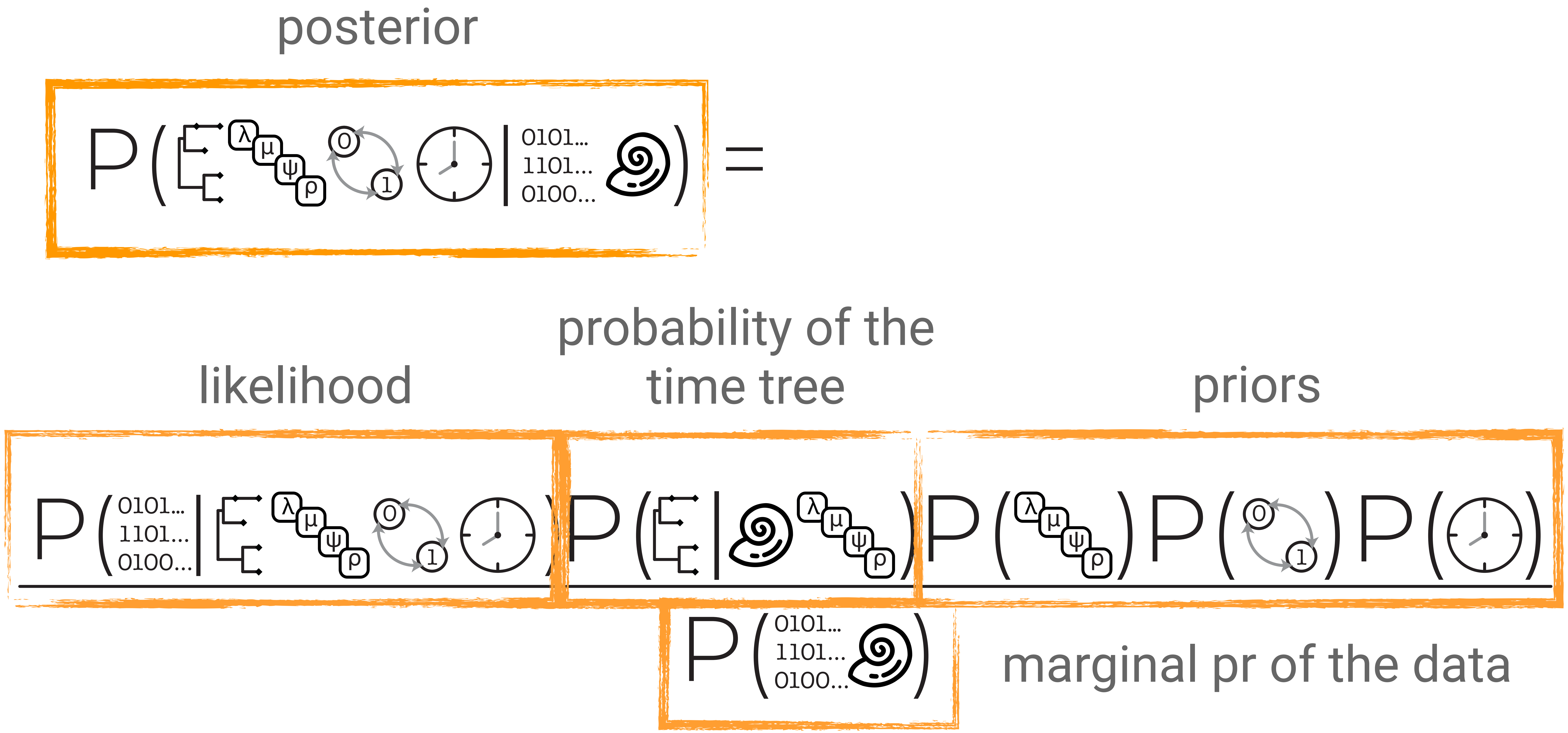


tree model

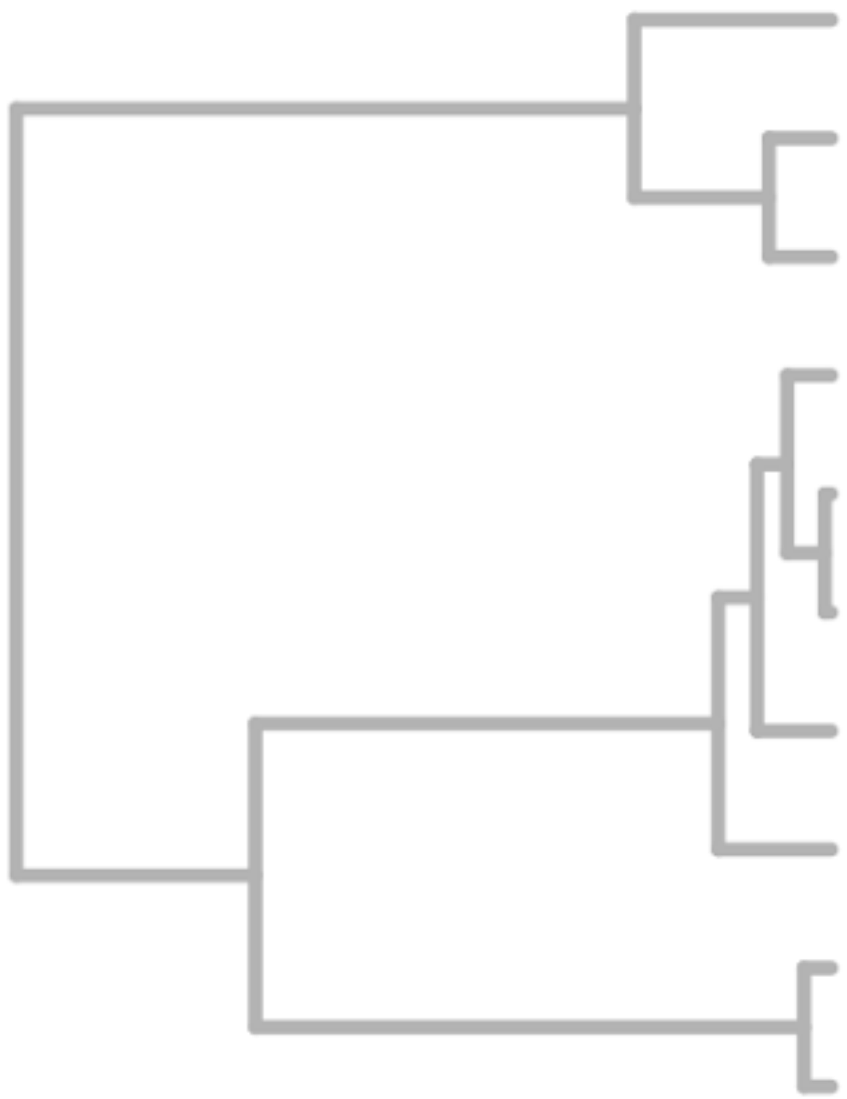


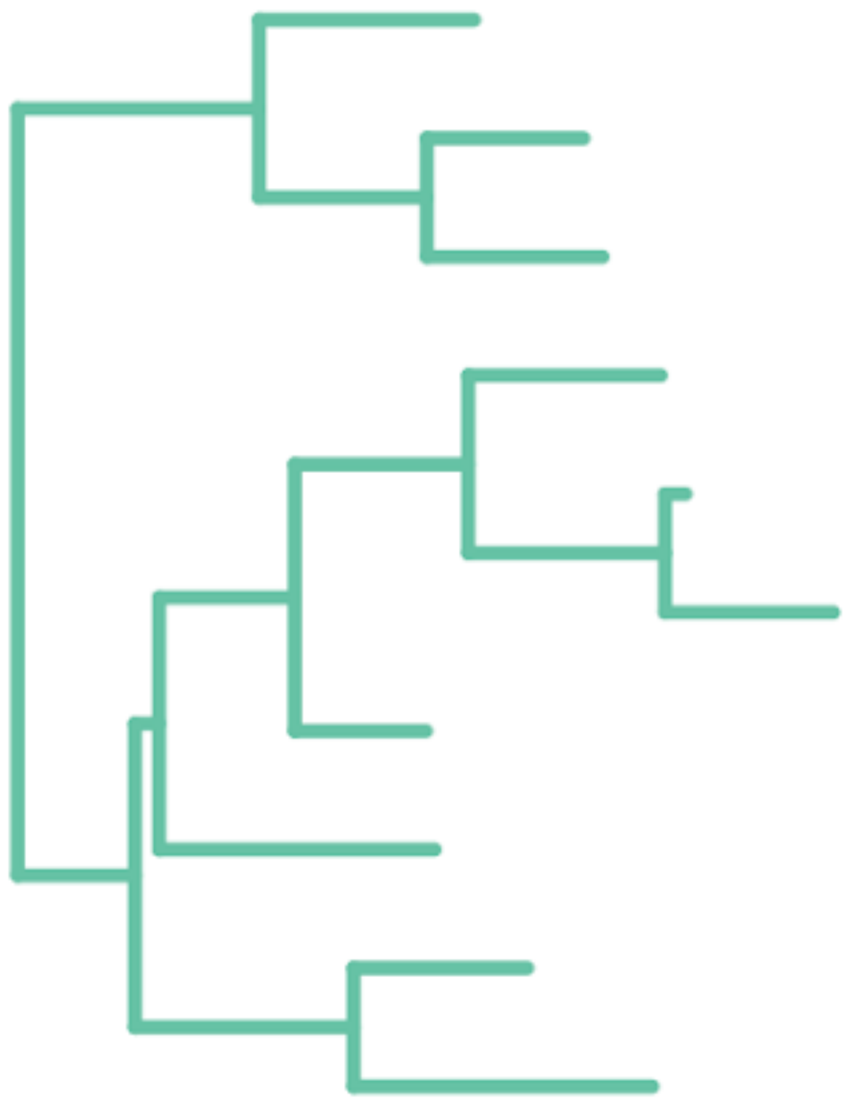
How have species originated, gone extinct and been sampled through time?

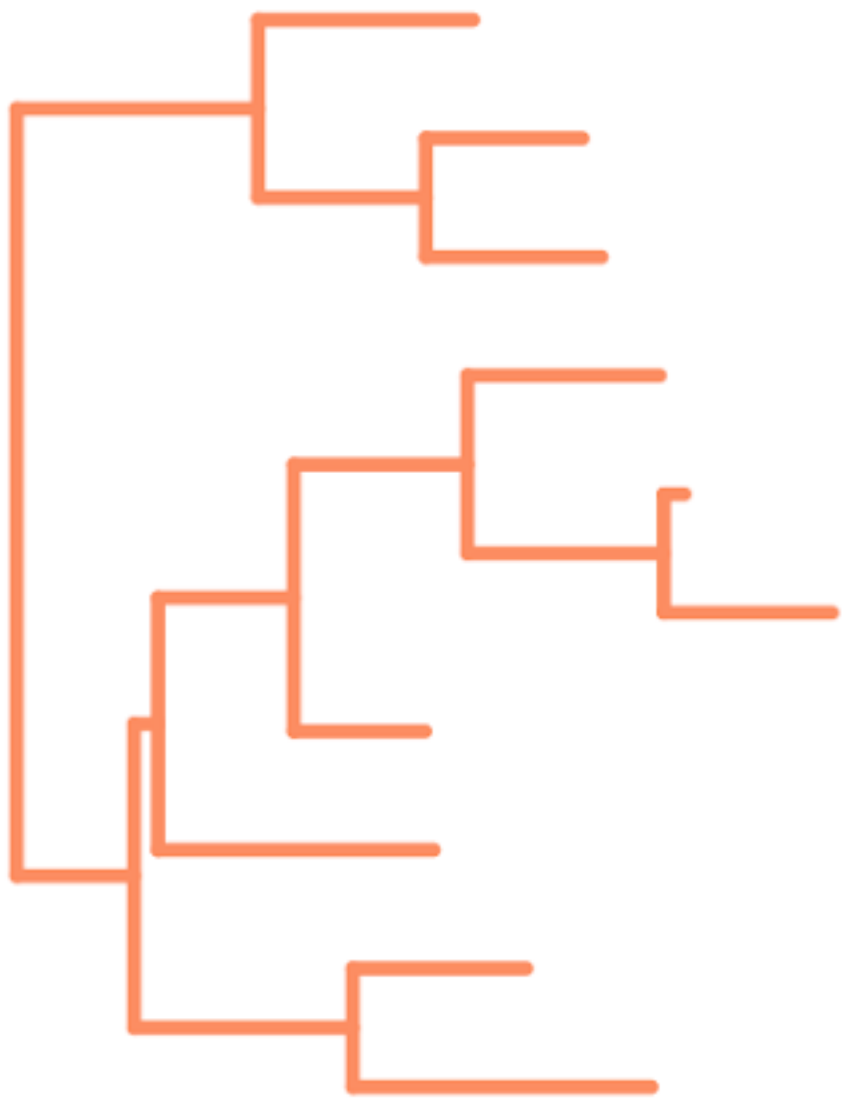
Bayesian divergence time estimation



Calculating the likelihood







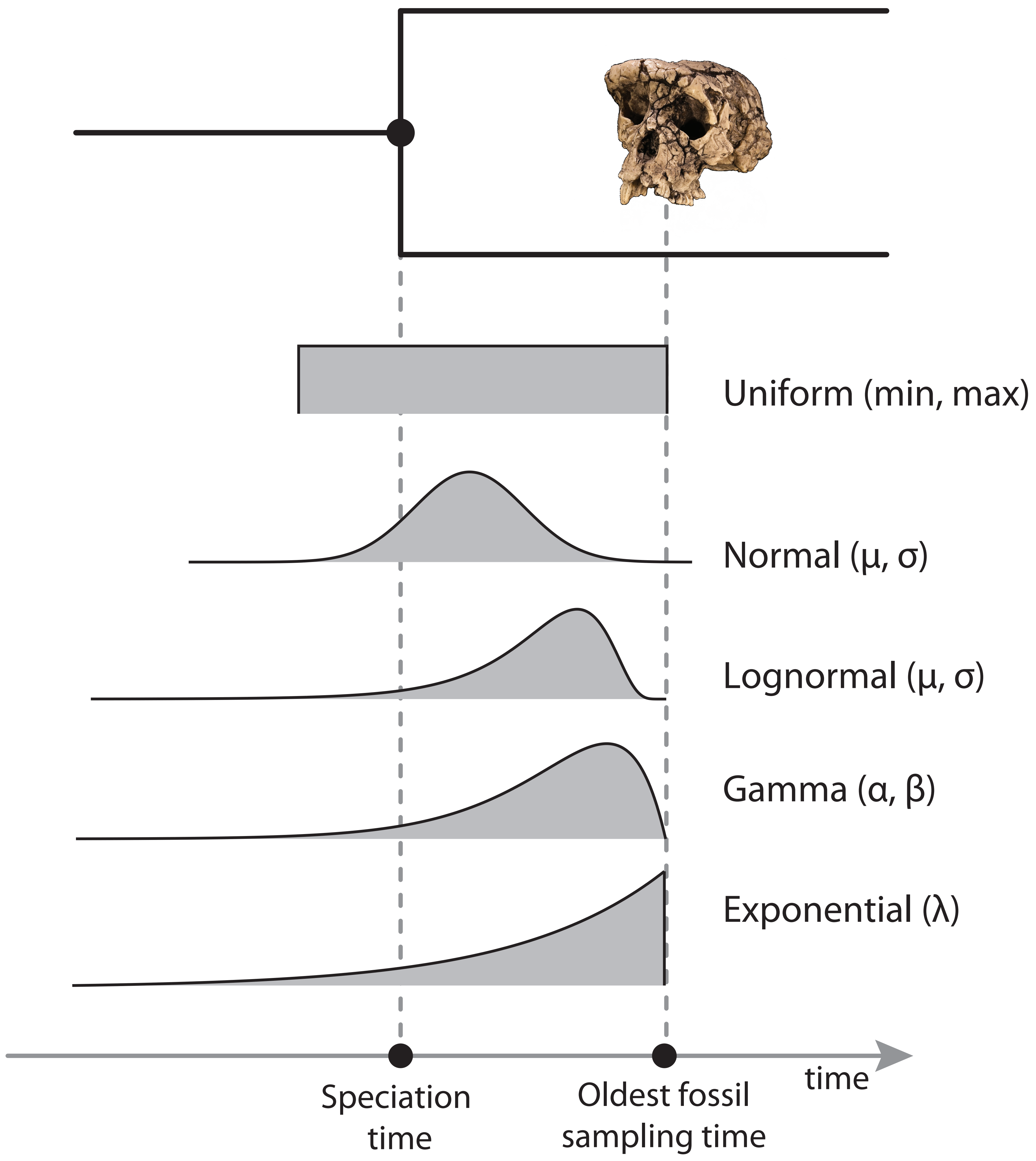
Time (year) X Clock rate (subs/site/year) = Genetic distance (subs/site)

Prior | Prior | Likelihood

Based on the calibration times we can estimate the rate over time

Once we have the rate we can transform evolutionary rates in genetic distance

Node dating



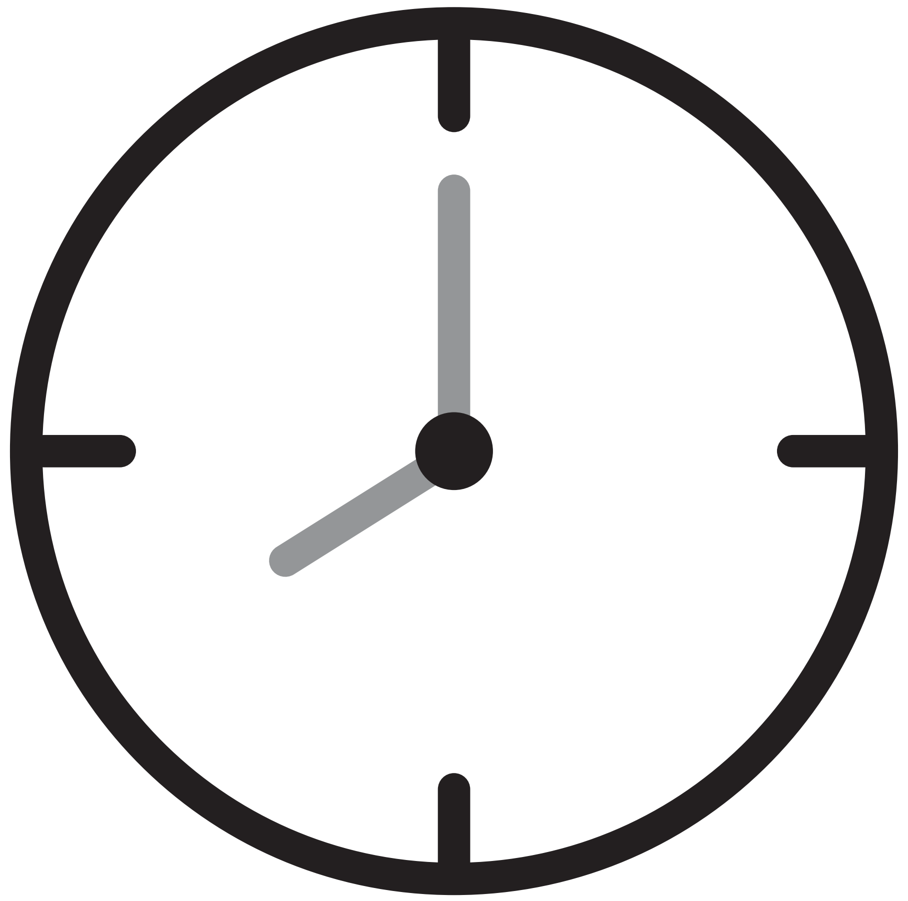
Extant taxa only in the tree

We can use a calibration density to constrain internal node ages

We typically use a birth-death process model to describe the tree generating process

Adapted from *Heath* 2012. Sys Bio

clock model

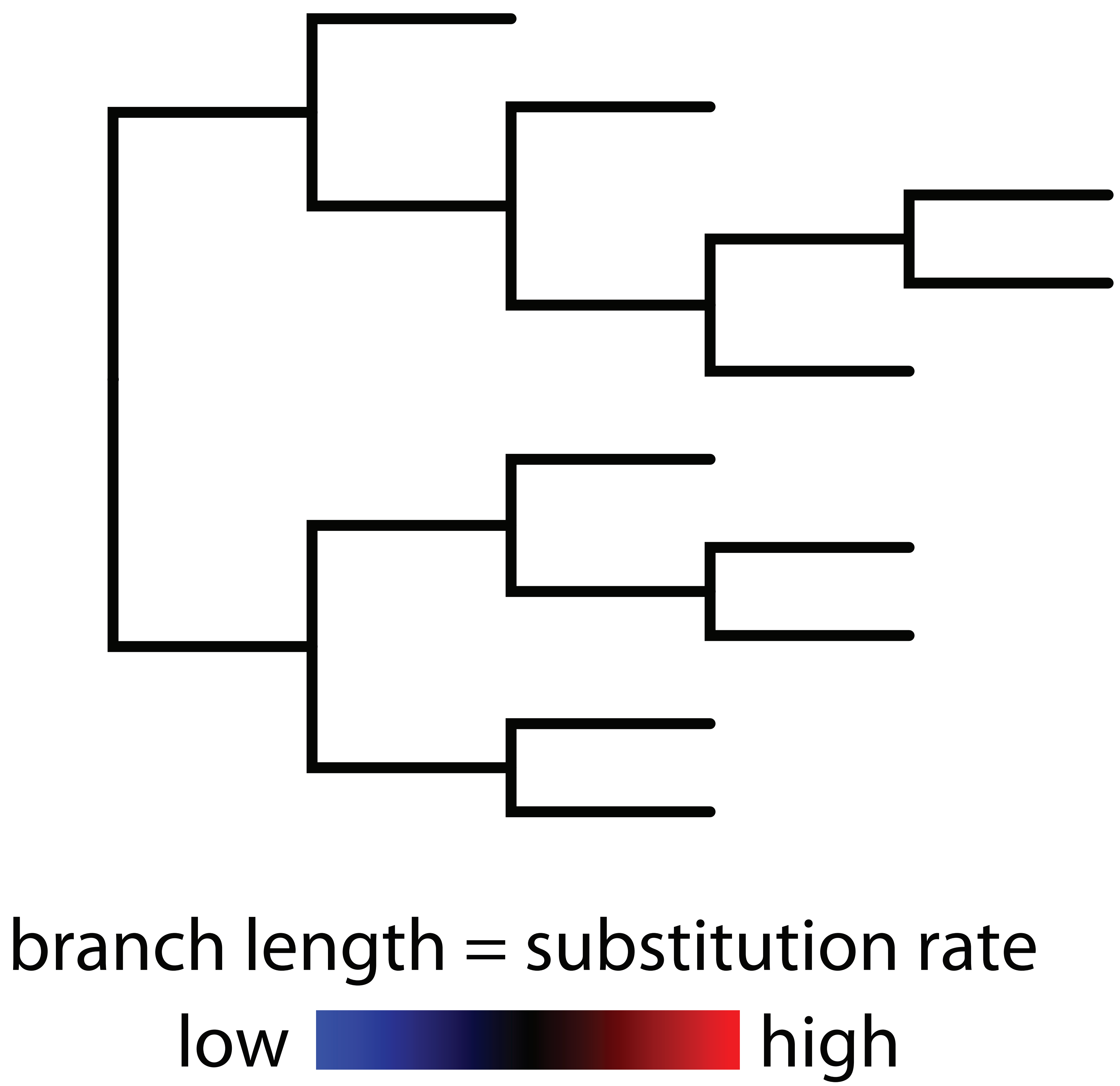


The clock model describes how evolutionary rates vary (or not) across the tree

The strict / constant molecular clock model

Assumptions

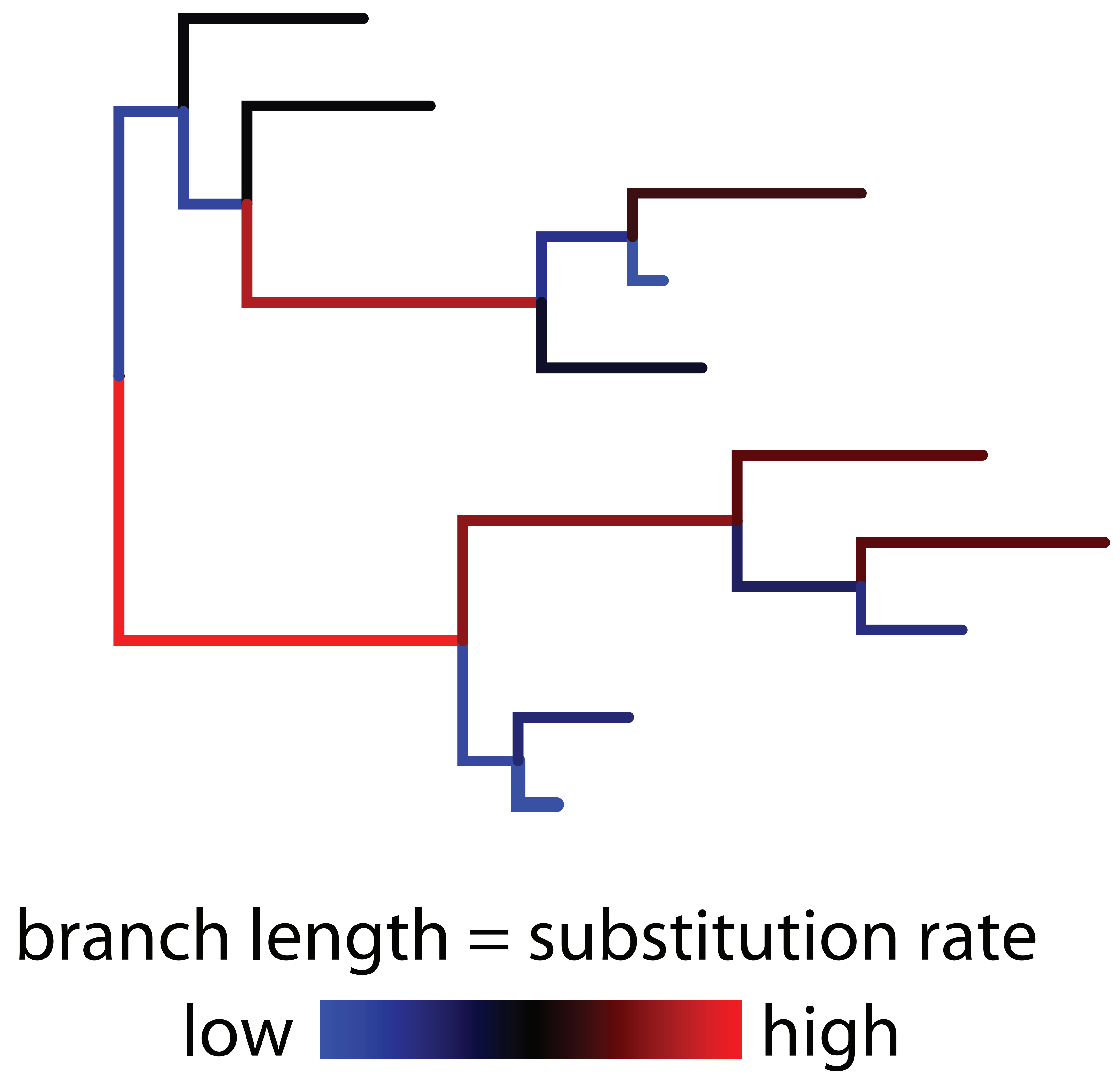
- The substitution rate is constant over time
- All lineages share the same rate



Relaxed clock models

Assumptions

- Lineage-specific rates
- The rate assigned to each branch is drawn from some underlying distribution



There are many different clock models

- Strict clock
- Uncorrelated or independent clock (= the favourite)
- Autocorrelated clock
- Local clocks
- Mixture models

The changing face of the molecular evolutionary clock. *Ho*. [2014](https://)
See also: *Warnock, Wright*. [2020](https://). Elements of Paleontology

Applications of node dating

Hypothesis testing: did the break up of Gondwana drive the radiation of cichlids?

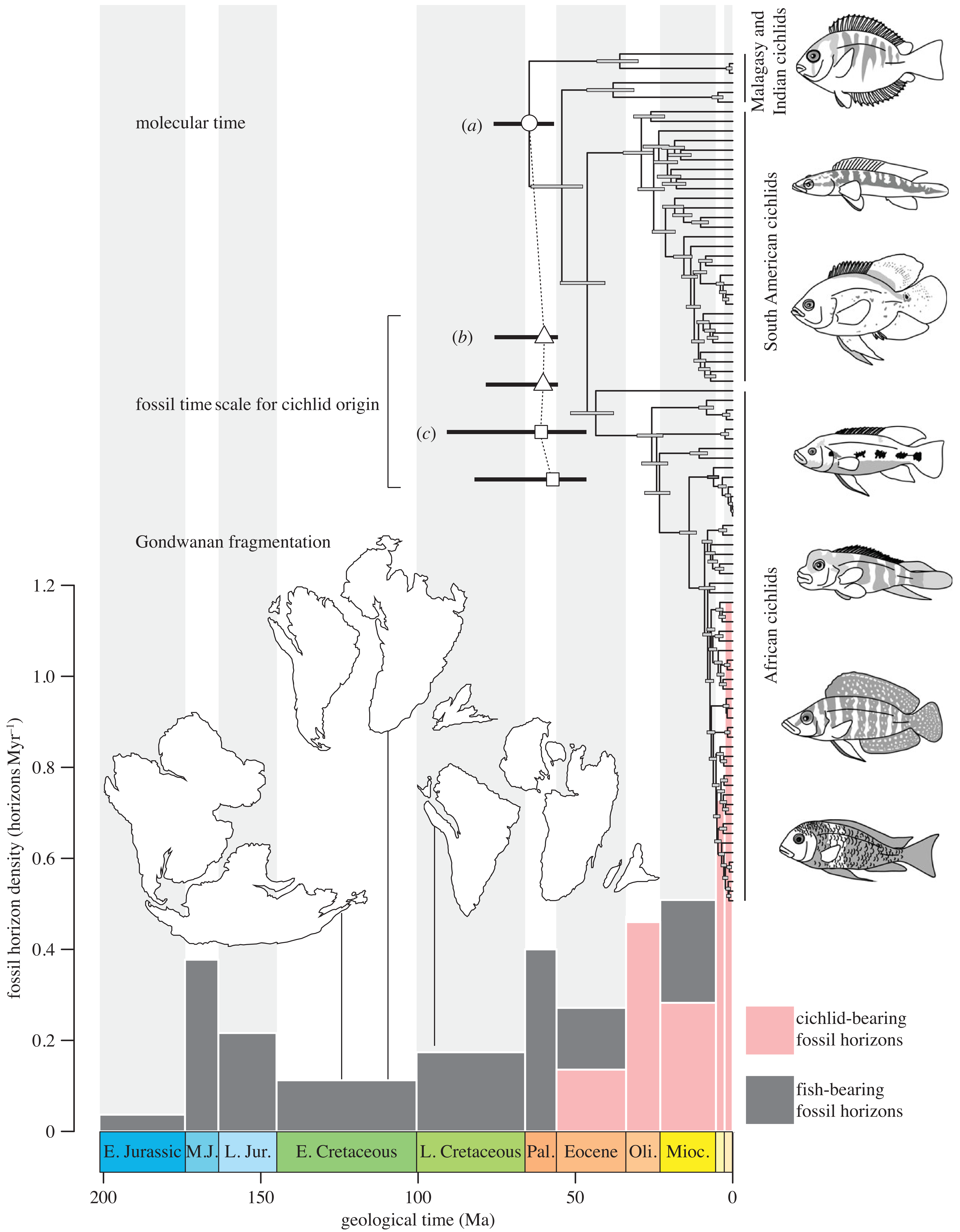
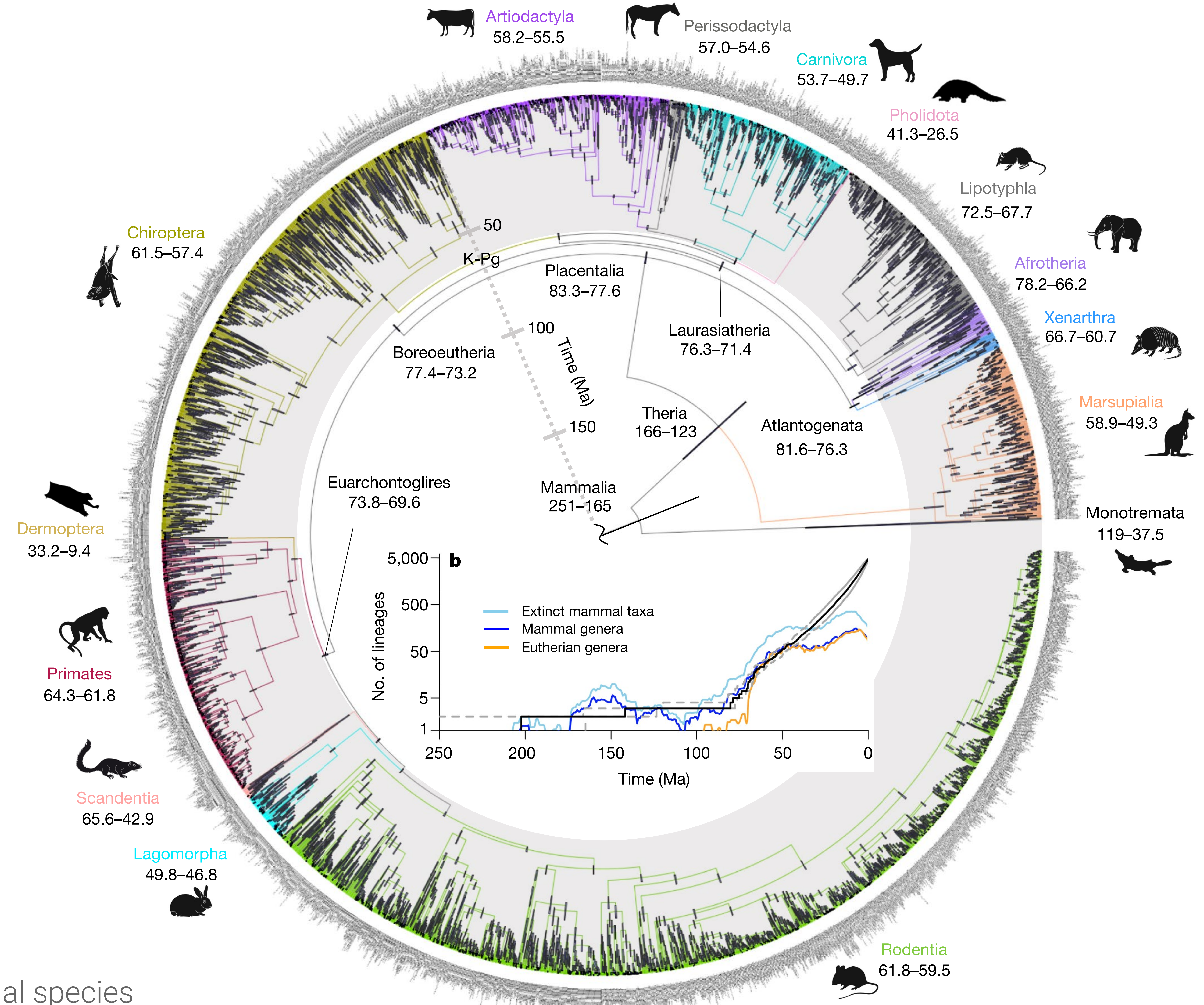


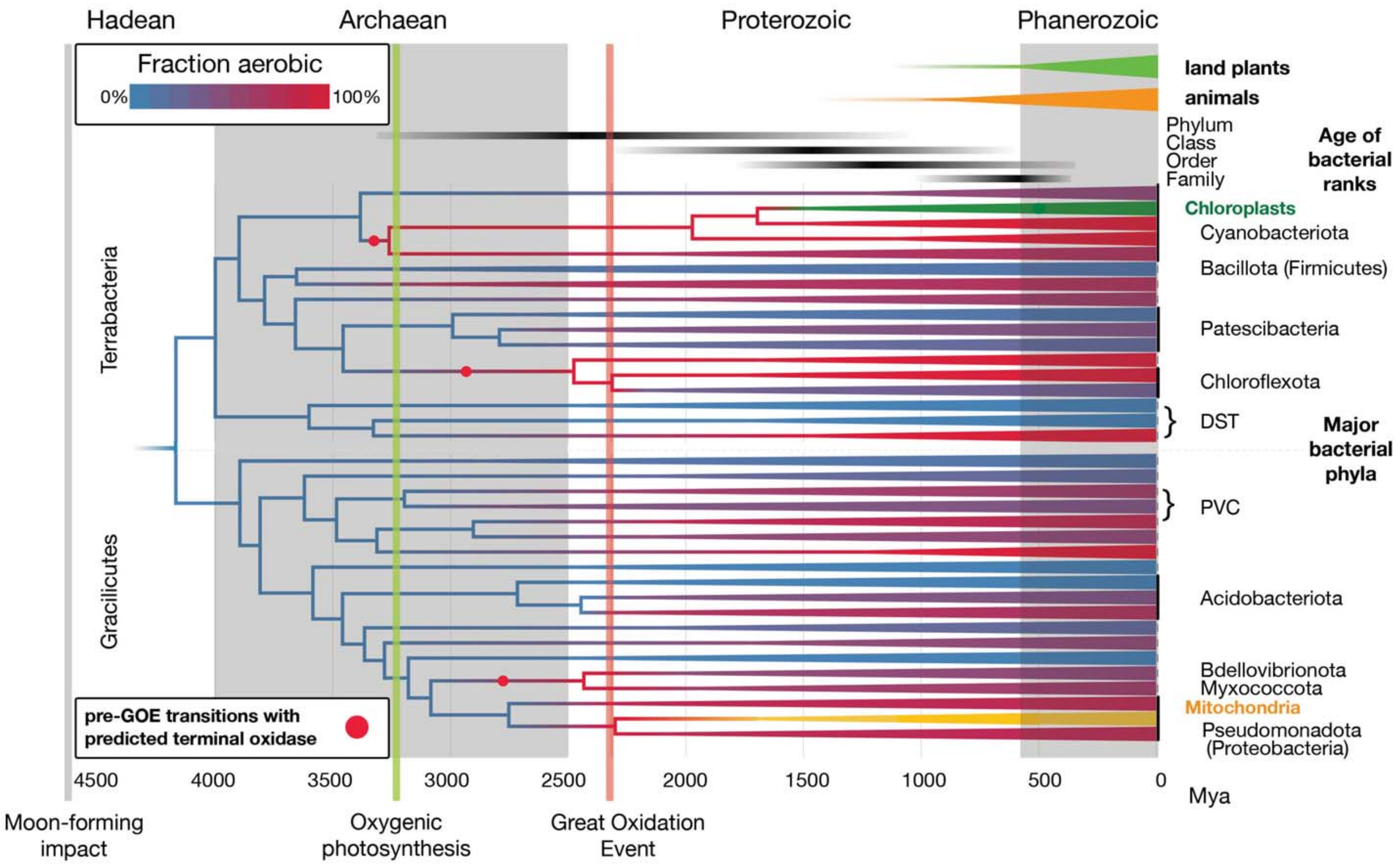
Image adapted from *Friedmann et al.* (2013)

Very large trees can only be time-calibrated using a node dating approach



Álvarez-Carretero et al. ([2021](#)) — 4,705 mammal species

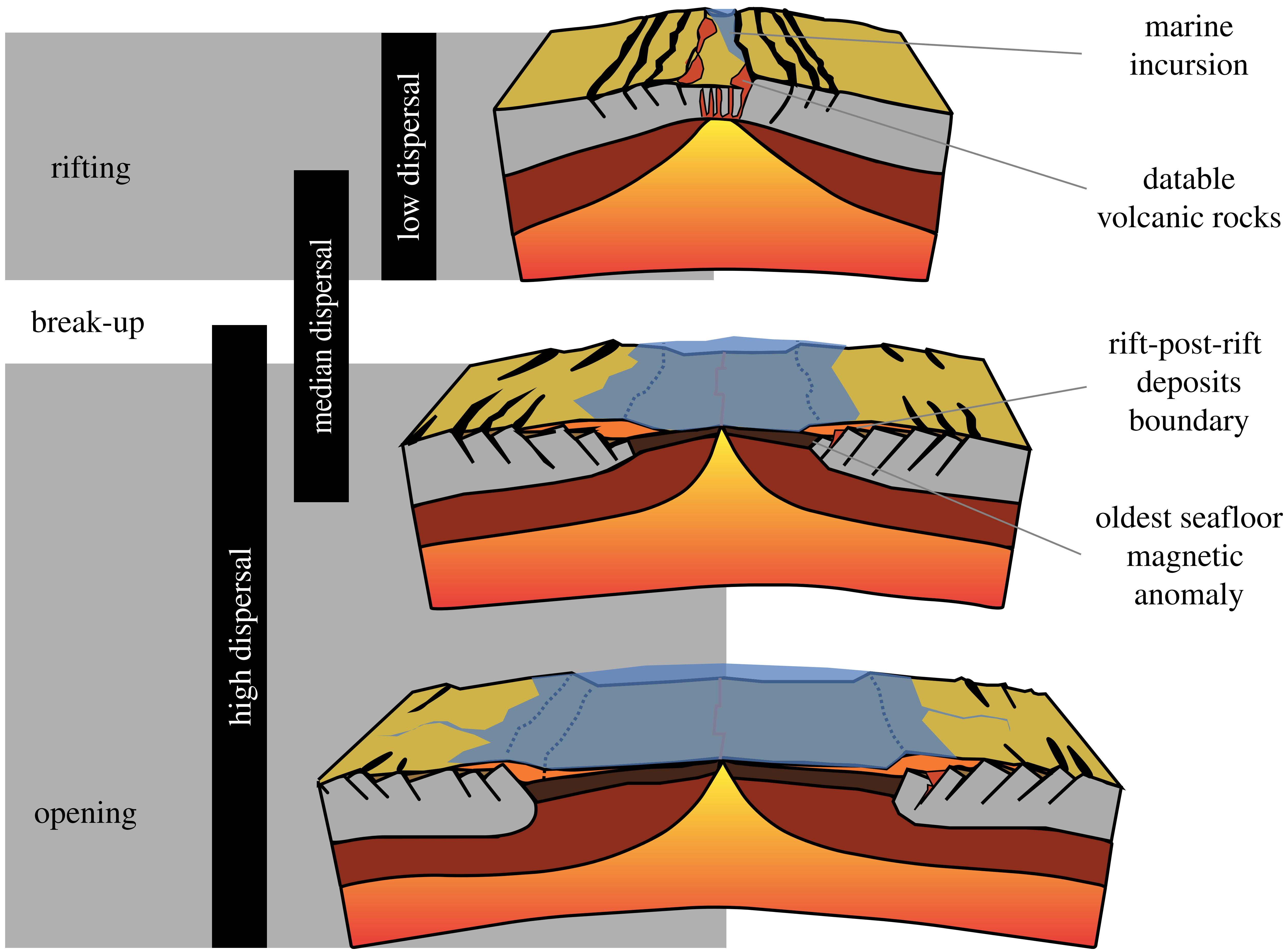
Dating with sparse calibration information



Davín et al. (2025)

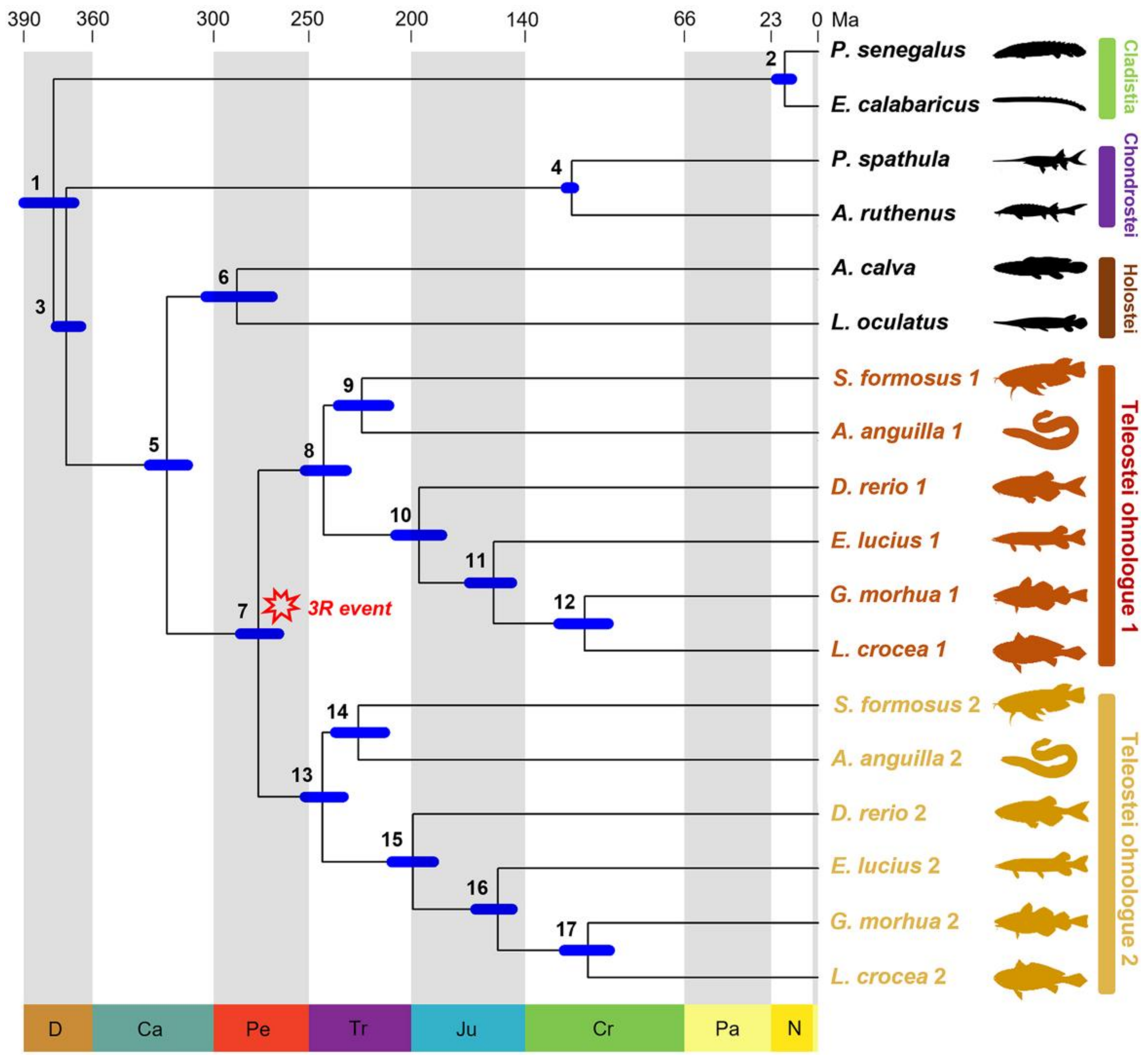
A geological timescale for bacterial evolution, calibrated using atmospheric oxygenation and the spread of aerobic metabolism

Biogeographic calibrations



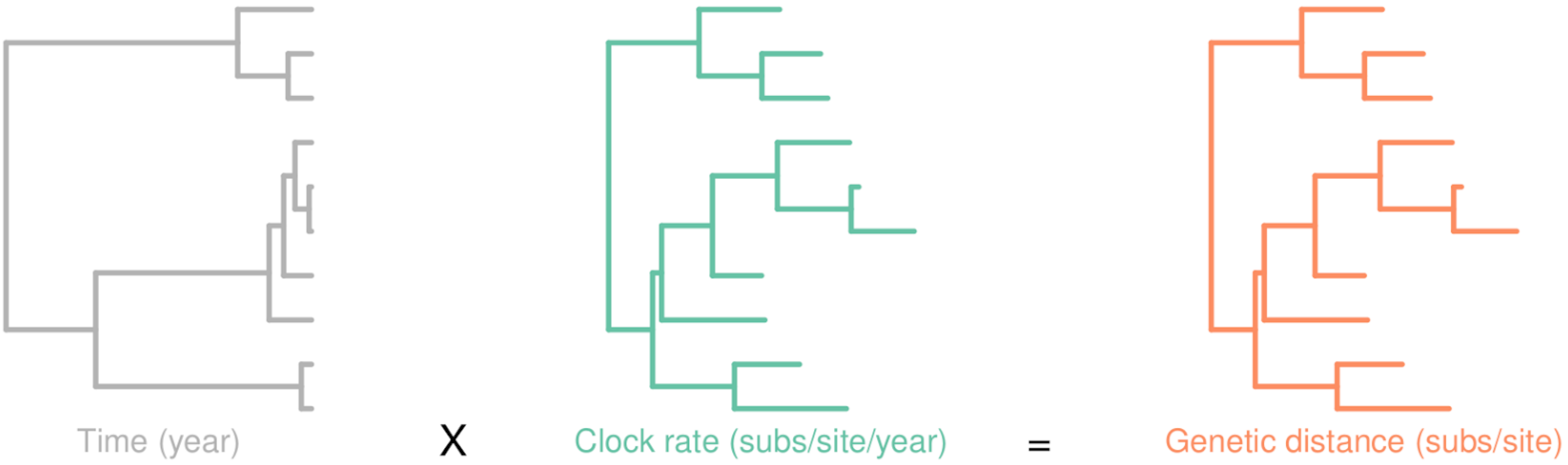
De Baets et al. (2016)
See also: Ho et al. (2015), *Landis* (2017, 2021), *Papadopoulou et al.* (2010)

Dating gene or genome duplication events



Missing reference

Times and rates are not fully identifiable!



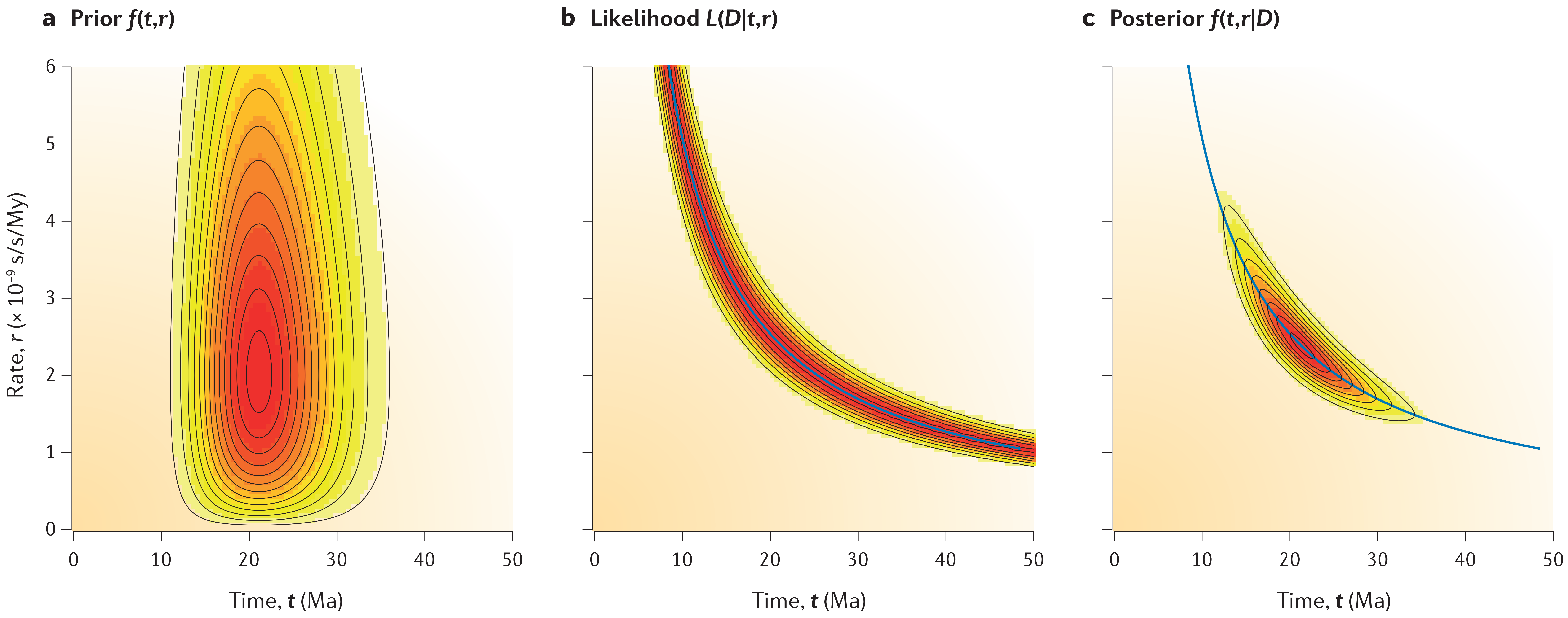
Prior

Prior

Likelihood

Slide adapted from Sebastian Duchene

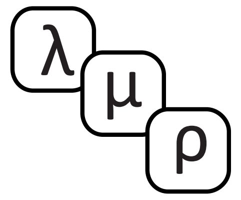
The priors will always influence the results



dos Reis et al. 2015. Nature Reviews Genetics

Intro to BEAST2

Phylogenetic inference — the old way

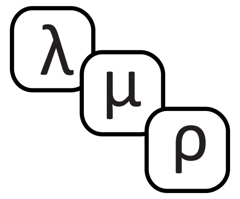


Model developers

What we might call a **“black box”** approach

Empirical researchers

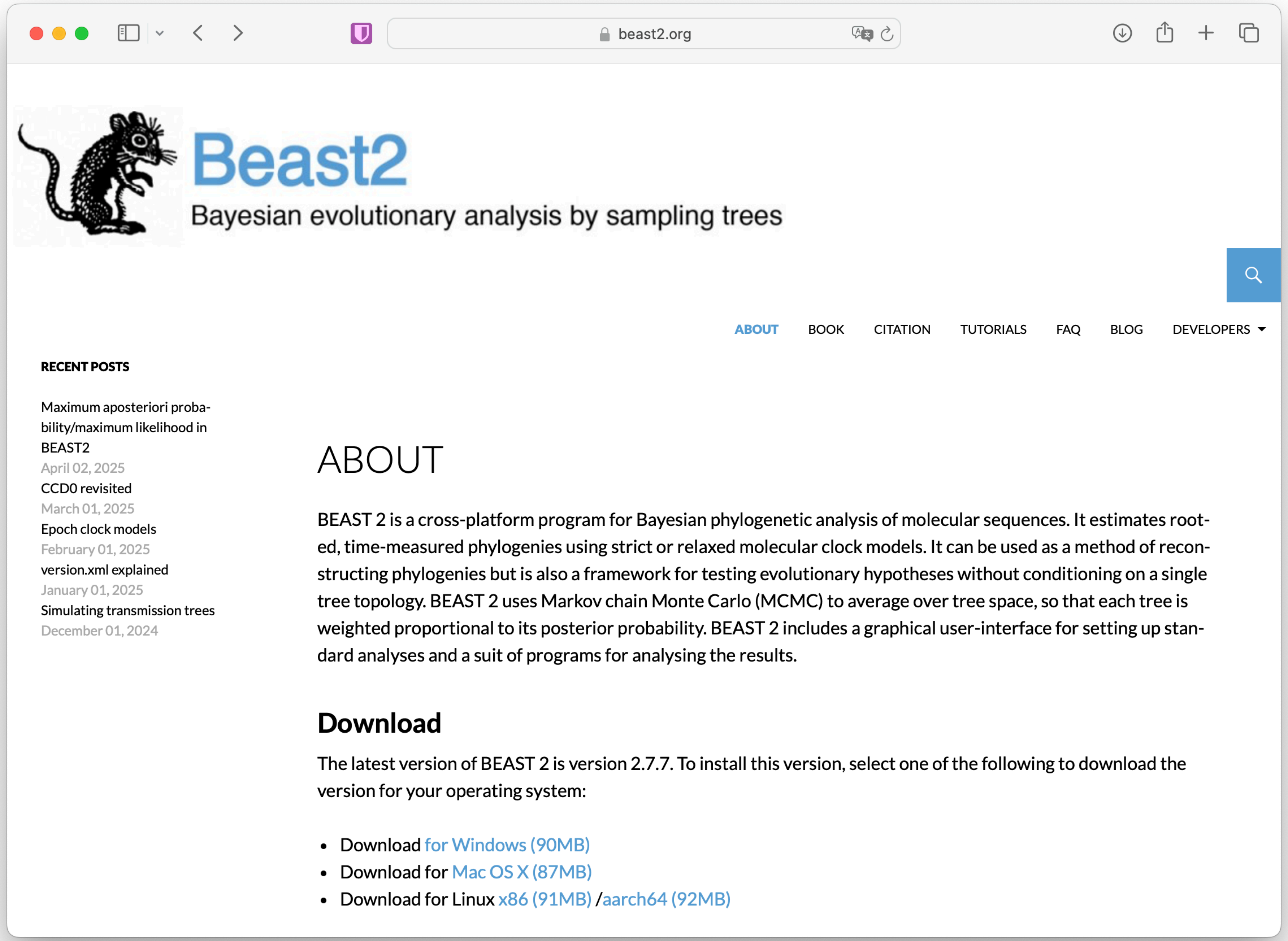
Phylogenetic inference — a better way?

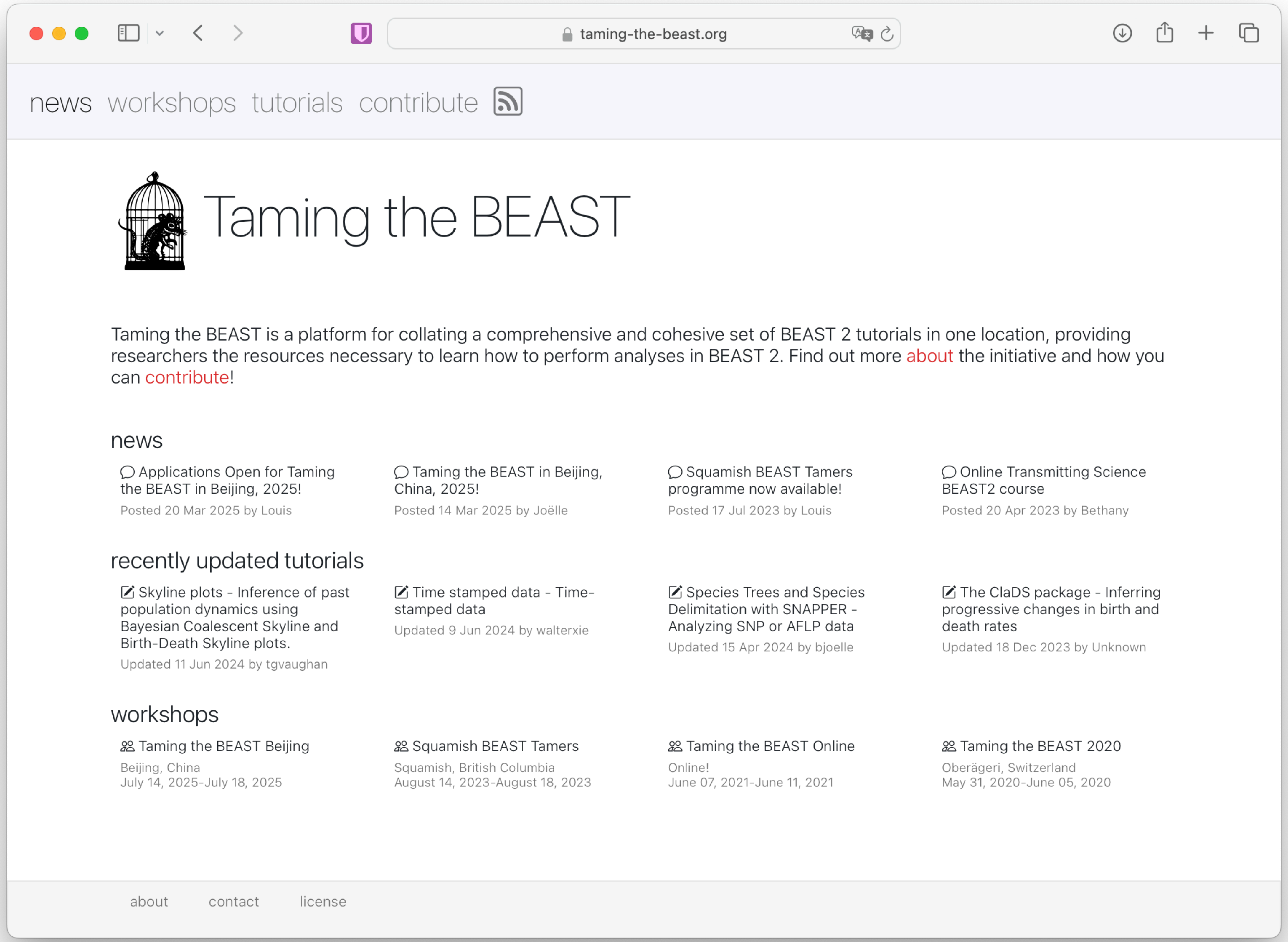


Model developers

Empirical researchers

The goal is to bring researchers with different expertise together, increase transparency, and do better research





www.beast2.org

taming-the-beast.org

BEAST2

Also designed with extendability and flexibility in mind

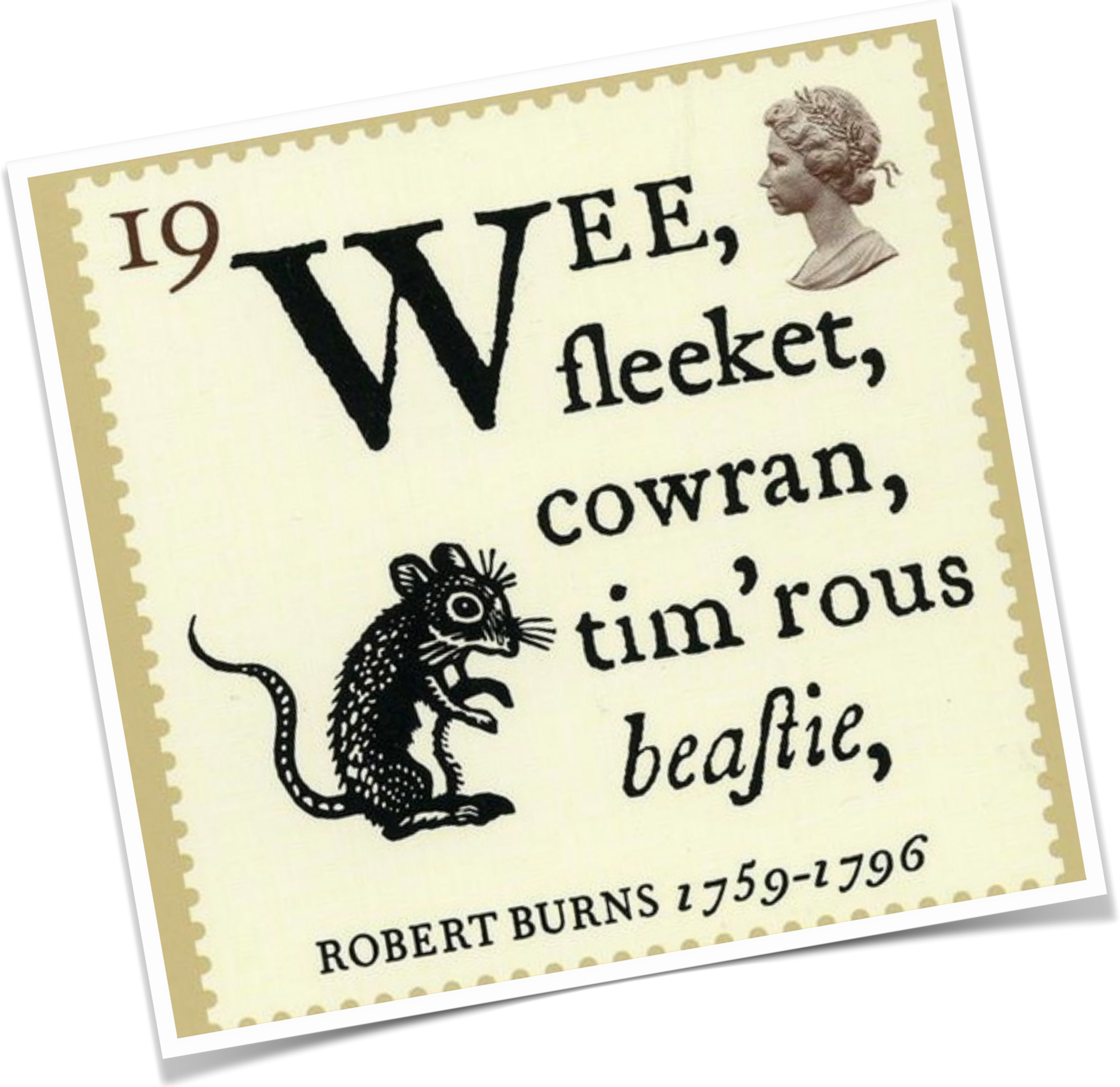
Also developed and supported by a large international team of developers

Has a suite of apps that can used to generate input files and analysis the output

www.beast2.org

Beast2

Bayesian evolutionary analysis by sampling trees



Scots poem - also the BEAST2 logo!

BEAST2 toolkit and work flow

Step 1. generate the xml file in **BEAUti**

Step 2. run your analysis in **BEAST**

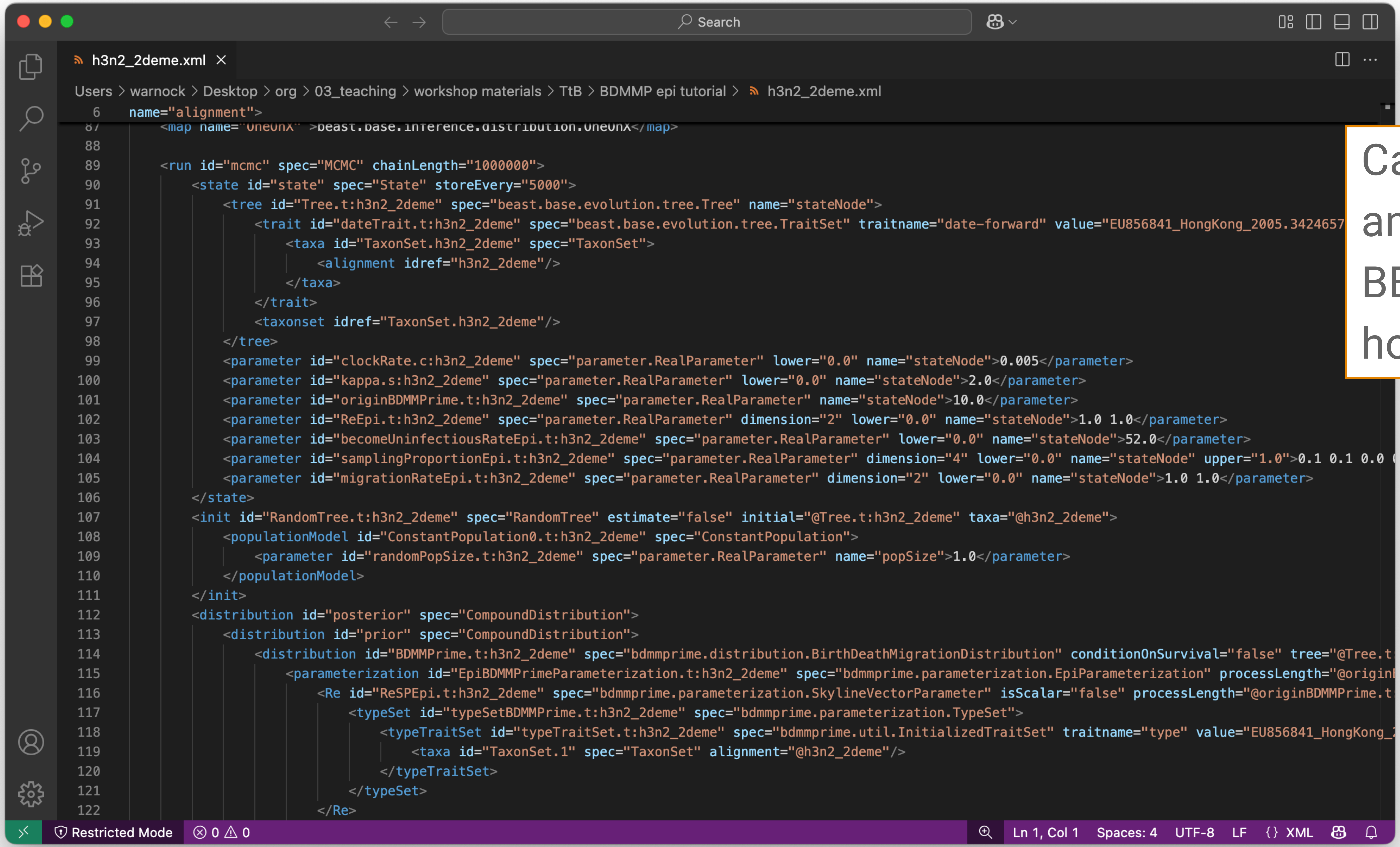
Step 3a. Examine you log files using **Tracer**

Step 3b. Generate a summary tree using **TreeAnnotator**

Step 4. Examine your summary tree in **FigTree**

Step… any other downstream analysis

BEAST2 input: the XML file



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            <trait id="dateTrait.t:h3n2_2deme" spec="beast.base.evolution.tree.TraitSet" traitname="date-forward" value="EU856841_HongKong_2005.3424657
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                </taxa>
            </trait>
            <taxonset idref="TaxonSet.h3n2_2deme"/>
        </tree>
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        </populationModel>
    </init>
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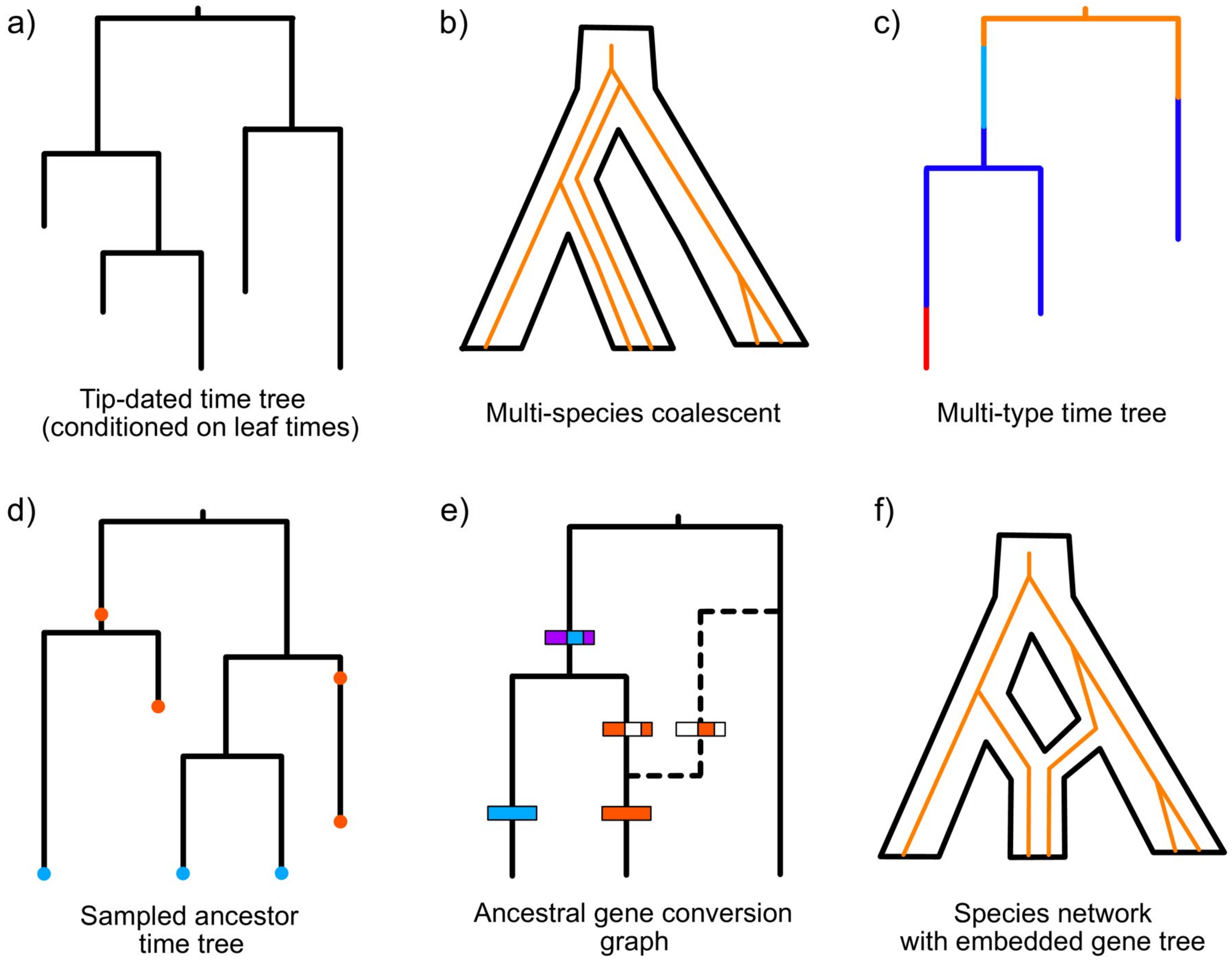
                                <taxa id="TaxonSet.1" spec="TaxonSet" alignment="@h3n2_2deme"/>
                            </typeTraitSet>
                        </typeSet>
                    </Re>
```

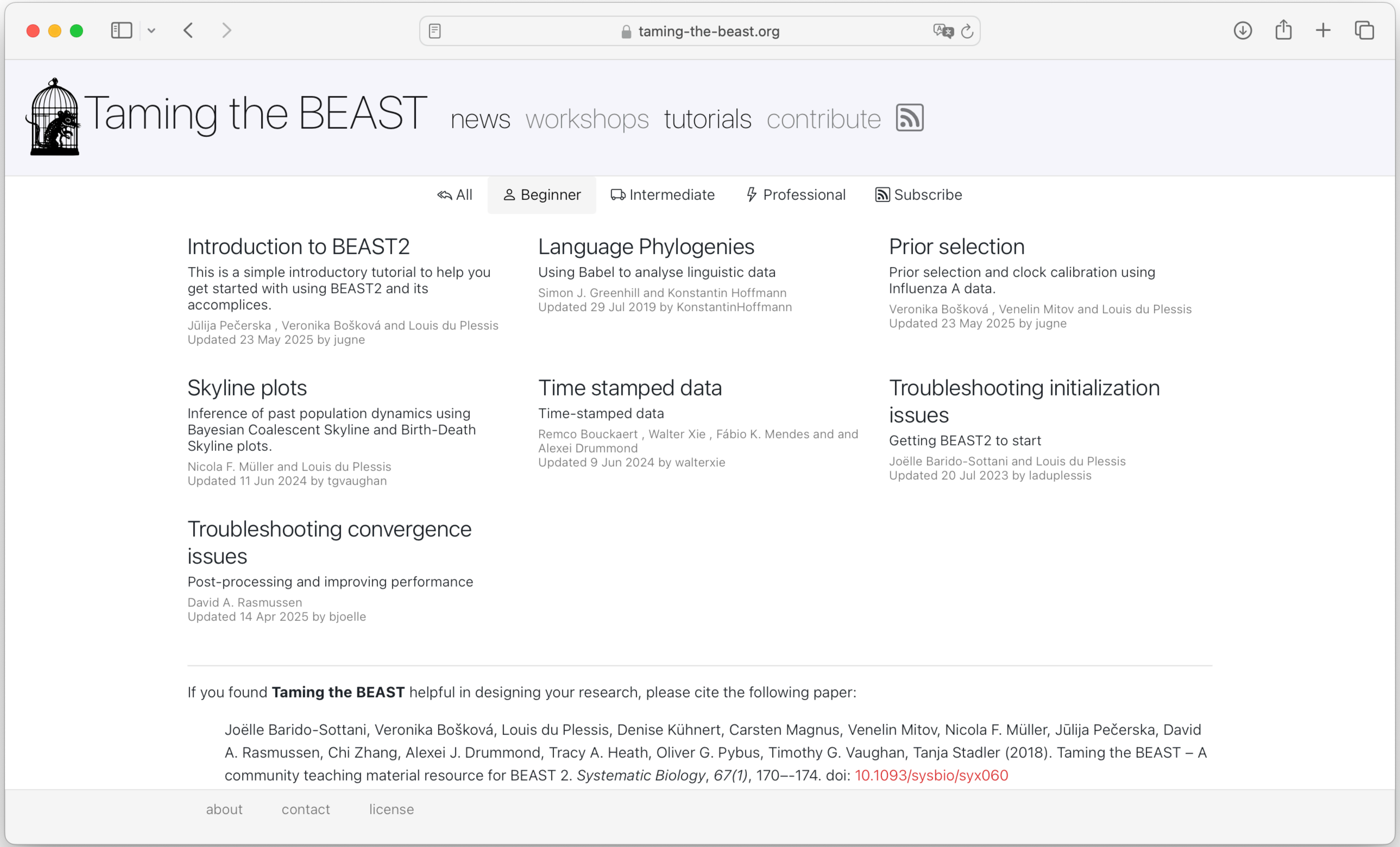
Caveat: if you want to use an option not available in BEAUti you have to learn how to edit the XML

A wide range of models and tree structures

Note all tree models in BEAST2 incorporate a temporal component



Bouckaert et al. ([2019](#)) *PLOS Comp Bio*



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> Joëlle Barido-Sottani, Veronika Bošková, Louis du Plessis, Denise Kühnert, Carsten Magnus, Venelin Mitov, Nicola F. Müller, Jūlija Pečerska, David A. Rasmussen, Chi Zhang, Alexei J. Drummond, Tracy A. Heath, Oliver G. Pybus, Timothy G. Vaughan, Tanja Stadler (2018). Taming the BEAST – A community teaching material resource for BEAST 2. *Systematic Biology*, *67(1)*, 170–-174. doi: 10.1093/sysbio/syx060

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Exercise

When you first open BEAUti, you might be asked if you want to install some packages - say yes!

You might encounter an error importing data into BEAUti the first time - just restart the program!